A single kinetic defect in a run-and-tumble ring: exact spectrum, exceptional points, and spectral-decay crossover

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Abstract. We determine how changing the tumbling rate at one site changes the spectrum and relaxation of a continuous-time run-and-tumble particle on an n-site ring. Reflection splits the state space into a protected sector and a defect-sensitive sector; a rank-one reduction then gives explicit finite polynomials and a scalar Green equation for the complete spectrum and Jordan structure, including the singular flat band at $\omega = 1$. On the infinite line, the same equation characterizes every localized mode outside the bulk spectrum, while a transfer argument excludes embedded ℓ^2 modes away from the explicitly characterized flat band. For fixed bulk rate $\omega > 0$, a slower defect—even one with $\omega_0 \asymp n^{-1}$—does not change the leading n^{-2} gap, whereas a faster defect changes the gap at order n^{-3}. When $\omega = \alpha/n$ and $\omega_0 = \beta/n$, the gap is instead of order n^{-1}; the controlling parity changes across $\alpha = 2\pi$, with an $n^{-3/2}$ correction at that scale for a faster defect. Finally, a second-order long-wave truncation restores the componentwise lattice-diffusion term absent from the leading telegraph model and reproduces the leading fixed-rate lattice decay.

Keywords. Run-and-tumble particle; kinetic defect; non-Hermitian spectrum; exceptional point; localized mode; spectral gap; telegraph equation.

Contents

1. Introduction

Persistent random walks describe motion with finite directional memory. The connection between correlated directional motion and the telegraph equation goes back to the analysis of Goldstein [1]. Continuum formulations of biased chemotactic motion arise, for example, in [2], while lattice run-and-tumble models retain the individual hopping and reversal events and connect directly to nonequilibrium particle systems [3]. For a homogeneous ring, the one-particle Fourier blocks used here coincide with the relative-coordinate blocks extracted in the two-particle analysis of [4]; that work gives their spectrum, exceptional points, and associated spectral-decay transition explicitly. For a continuum relative-coordinate model with thermal diffusion, [5, Secs. 3.2–3.3] analyzes relaxation, four symmetry sectors, and the low-lying spectrum on a ring. Section 2 records the precise convention used below.

This paper isolates the spectral effect of one fixed tumbling-rate defect. The fixed-site geometry preserves a reflection symmetry, and that symmetry leaves one parity sector unchanged while reducing the other to a scalar rank-one equation. The resulting analysis determines the finite-ring spectrum and Jordan structure, classifies infinite-line localized states, and identifies which parity controls the large-ring decay rate in two scaling regimes.

Spatial inhomogeneity has been studied through position-dependent speed [6], quenched disorder in speed and tumbling [7, Secs. II.A.2–II.A.3 and II.B.2–II.B.3], saturating space-dependent rates [8, Sec. II, Eqs. (3)–(4)], and direction- and position-dependent reversal rates [9, Secs. 2–4]. A kinetic description with signal-modulated tumbling and its relaxation dynamics is developed in [10]. These works emphasize probability laws, steady states, first-passage behavior, or relaxation in extended media. A continuum particle with tumbling localized at the two boundaries of an interval is solved in [11, Sec. III, Eqs. (16) and (29)–(30)]; its local reaction terms yield time-dependent densities rather than the complete spectrum of one fixed lattice-site defect. Impurity-activated reversals and moving tumble sites also occur in interacting periodic exclusion processes [12, Secs. V–VII], [13, Secs. II–IV]; their focus is driven clustering and many-particle steady behavior rather than the full one-particle spectrum of a fixed tumbling-rate defect.

Related lattice-impurity methods include persistent walks with random asymmetric transmittances [14], Chebyshev propagators and partially absorbing defects for confined nonpersistent walks [15, Secs. VI–VII and Apps. C and E], and the classical difference-equation construction of isolated impurity levels [16]. Green-function reduction of a point perturbation has also been used for non-Hermitian lattices [17, Secs. II–III and Eqs. (11)–(15)]. Finite-size impurity spectra and impurity-induced size-dependent localization have been studied in nonreciprocal chains [18]. Complex-impurity localization and exceptional points in otherwise Hermitian reciprocal lattices are treated in [19]. Those tight-binding problems concern on-site potentials, boundary sensitivity, and skin or bound-state localization. Here the local perturbation acts instead on the tumbling channel of a stochastic generator.

Rank-one determinant and resolvent identities are the finite-dimensional mechanism behind the scalar reduction; an early inverse-update formula appears in [20, pp. 124–127]. For scalar second-order difference operators, transfer matrices, Green kernels, and periodic monodromy are developed in [21, Secs. 1.1–1.2 and 7.1]. The operator here is instead a two-component nonnormal stochastic generator, so the parity and Jordan analysis is carried out directly. Chapters 14–15 of [22] review the distinction between asymptotic spectral decay and transient behavior for nonnormal matrix exponentials.

Here and below, an exact spectral description means that the eigenvalues and their Jordan multiplicities are characterized by explicit finite polynomials or secular equations; it does not mean that the roots of a general polynomial are expressed in radicals.

Table 1 maps the principal conclusions to their exact statements. In particular, the gap theorems combine local first-pole expansions with global right-edge root exhaustion.

Object or regime	Main conclusion	Location
Finite ring	Characteristic equations and parity factorization	Theorems 3.3, 3.4, and 4.4
Exceptional points	Jordan classification, including the flat band at $\omega = 1$	Theorems 4.8–4.9
Infinite line	Localized modes outside the bulk spectrum, embedded-spectrum exclusion, and compact modes	Theorem 5.4 and Propositions 5.3 and 5.6
Fixed $\omega > 0$	Protected/affected gap selection and fixed-rate expansion	Theorem 6.4
Bulk-critical rates	Bulk-critical gap and parity-selection phase diagram	Theorem 7.1
Continuum models	Telegraph error, lattice diffusion, and point interaction	Theorem 8.3 and Proposition 8.5

Table 1: Map of the principal results.

Section 2 defines the model; Sections 3– 5 give the finite- and infinite-volume spectral structure. Sections 6–7 determine the two gap regimes, and Section 8 gives the continuum comparisons and point-defect analogue.

2. Model and notation

Definition 2.1 (Defective run-and-tumble generator)**.** Let $n \geq 2$, let $\omega > 0$, and let $\omega_0 \geq 0$. Sites are $j \in \mathbb{Z}/n\mathbb{Z}$, orientations are $\sigma \in \{+,-\}$, and

$$\omega_j = \begin{cases} \omega_0, & j = 0, \\ \omega, & j \neq 0. \end{cases}$$

The forward generator $L = L_n(\omega, \omega_0)$ is

$$(Lu)_j^+ = u_{j-1}^+ - (1+\omega_j)u_j^+ + \omega_j u_j^-, \tag{2.1}$$

$$(Lu)_j^- = u_{j+1}^- - (1+\omega_j)u_j^- + \omega_j u_j^+. \tag{2.2}$$

Thus hopping in the current orientation has rate one and tumbling has rate ω_j. We regard $u = (u_j^+, u_j^-)_{j\in\mathbb{Z}/n\mathbb{Z}}$ as a column vector in $\mathbb{C}^{2n}$ and write $e_{j,\sigma}$ for the corresponding coordinate vector. With this convention, the columns of L sum to zero, so $\partial_t u = Lu$ preserves the total mass of a probability vector. Put

$$\delta = \omega_0 - \omega, \qquad q_k = \frac{2\pi k}{n}, \qquad c_k = \cos q_k, \qquad s_k = \sin q_k.$$

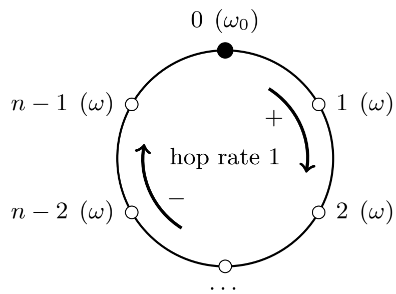


Figure 1: The run-and-tumble ring. Tumbling exchanges the two orientations at rate ω_j; the marked site has rate ω_0 and every other site has rate ω.

Equations (2.1)–(2.2) use the same one-particle convention as equations (6)–(10) of [4]. That paper records the homogeneous eigenvalues in its equation (10), the homogeneous exceptional points in equation (13), and the first spatial-mode decay rate in equations (20)–(21).

Lemma 2.2 *(Rank-one defect)***.** Let $L^{(0)} = L_n(\omega, \omega)$. If $d = e_{0,+} - e_{0,-}$, then

$$L = L^{(0)} - \delta\, dd^{\mathsf{T}}.$$

In particular, the perturbation has rank one when $\delta \neq 0$. Here $^{\mathsf{T}}$ is the matrix transpose in the displayed coordinate basis; Hermitian inner products are written with angle brackets below.

Proof. At site zero, changing ω to ω_0 adds

$$\delta \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} = -\delta \begin{pmatrix} 1 \\ -1 \end{pmatrix} \begin{pmatrix} 1 & -1 \end{pmatrix}.$$

Every other matrix entry is unchanged. ∎

Notation guide. The scalar δ is the defect contrast and d is the polarization vector at the defect; distances are always written explicitly below. The unadorned R denotes kinetic reflection, whereas decorated symbols such as R_n and $R^{\pm}_{\ell,n}$ denote pole residues. The finite and infinite polarization Green functions are G_n and g_∞, respectively. In the gap sections, $q = 2\pi/n$ is the first finite-ring wave number and $q_\star = 2\pi$ is its scaled value.

3. Equivalent characteristic formulations

Definition 3.1 (Homogeneous factors)**.** For $\lambda \in \mathbb{C}$, set

$$\begin{aligned} a &= \lambda + 1 + \omega, && (3.1)\\ x &= \frac{a^2 + 1 - \omega^2}{2a}, && (3.2)\\ D_k(\lambda) &= (\lambda + 1 - c_k)(\lambda + 1 + 2\omega - c_k) + s_k^2 = a^2 + 1 - \omega^2 - 2ac_k. && (3.3)\end{aligned}$$

Although x is undefined at $a = 0$, each expression below containing $a^n T_n(x)$ is understood by its unique polynomial continuation.

Lemma 3.2 *(Defect-free characteristic polynomial)***.** The defect-free characteristic polynomial is

$$\begin{aligned} P_n^{(0)}(\lambda;\omega) &= \det(\lambda I - L^{(0)})\\ &= \prod_{k=0}^{n-1} D_k(\lambda) && (3.4)\\ &= 2a^n\big(T_n(x) - 1\big), && (3.5)\end{aligned}$$

where T_n is the Chebyshev polynomial of the first kind. The two roots of D_k are

$$\lambda_k^{\pm} = c_k - 1 - \omega \pm \sqrt{\omega^2 - s_k^2}.$$

Proof. Fourier transformation gives the block

$$W_k = \begin{pmatrix} e^{\mathrm{i}q_k} - 1 - \omega & \omega \\ \omega & e^{-\mathrm{i}q_k} - 1 - \omega \end{pmatrix}.$$

Its characteristic polynomial is D_k, which proves (3.4) and the displayed roots. Since $D_k = 2a(x - c_k)$ and

$$\prod_{k=0}^{n-1}(x - c_k) = 2^{1-n}\big(T_n(x) - 1\big),$$

equation (3.5) follows. ∎

Theorem 3.3 *(Exact finite-ring characteristic polynomial)***.** For every $n \geq 2$, $\omega > 0$, and $\omega_0 \geq 0$,

$$\begin{aligned} P_n(\lambda;\omega,\omega_0) &:= \det(\lambda I - L_n(\omega,\omega_0)) && (3.6)\\ &= P_n^{(0)}(\lambda;\omega) + \frac{\delta}{n}\,\partial_\omega P_n^{(0)}(\lambda;\omega) && (3.7)\\ &= P_n^{(0)}(\lambda;\omega)\left[1 + \frac{2\delta}{n}\sum_{k=0}^{n-1}\frac{\lambda + 1 - c_k}{D_k(\lambda)}\right]. && (3.8)\end{aligned}$$

Equation (3.8) is interpreted by polynomial continuation at a homogeneous eigenvalue. Equivalently,

$$P_n \;=\; 2a^n(T_n(x)-1)+2\delta a^{n-1}(T_n(x)-1)+\delta a^{n-2}\lambda(\lambda+2)U_{n-1}(x), \tag{3.9}$$

where U_{n-1} is the Chebyshev polynomial of the second kind and the right-hand side again denotes its polynomial continuation.

Proof. Let $B \;=\; \lambda I - L^{(0)}$. Lemma 2.2 and the matrix determinant lemma give

$$\det(\lambda I - L) \;=\; \det B\bigl(1+\delta d^{\mathsf{T}}B^{-1}d\bigr).$$

In Fourier space,

$$d^{\mathsf{T}}B^{-1}d \;=\; \frac{1}{n}\sum_{k=0}^{n-1}\frac{2(\lambda+1-c_k)}{D_k(\lambda)},$$

which proves (3.8). Direct differentiation of (3.4) gives

$$\partial_\omega P_n^{(0)} \;=\; P_n^{(0)}\sum_k \frac{2(\lambda+1-c_k)}{D_k},$$

and hence (3.7). Finally,

$$\partial_\omega x \;=\; \frac{\lambda(\lambda+2)}{2a^2}, \qquad T_n'(x) \;=\; nU_{n-1}(x),$$

and differentiating (3.5) proves (3.9). ■

Theorem 3.4 *(Transfer-matrix quantization)***.** For a local rate $r \;\geq\; 0$, define

$$T_r(\lambda) \;=\; \frac{1}{\lambda+1+r}\begin{pmatrix}1 & r\\ -r & (\lambda+1+r)^2-r^2\end{pmatrix}. \tag{3.10}$$

Then $\det T_r \;=\; 1$, and, away from the displayed denominators,

$$P_n(\lambda;\omega,\omega_0) \;=\; a^{n-1}(\lambda+1+\omega_0)\left\{\operatorname{tr}\bigl[T_\omega(\lambda)^{n-1}T_{\omega_0}(\lambda)\bigr]-2\right\}. \tag{3.11}$$

After multiplication by the denominators, this identity holds for every λ by polynomial continuation. When $a \;\neq\; 0$ and $\lambda+1+\omega_0 \;\neq\; 0$, an eigenvalue is equivalently characterized by

$$\operatorname{tr}\bigl[T_\omega(\lambda)^{n-1}T_{\omega_0}(\lambda)\bigr] \;=\; 2. \tag{3.12}$$

At a zero of either denominator, the cleared polynomial identity (3.11), rather than the undefined trace in (3.12), is the eigenvalue criterion. Moreover, if $t \;=\; \operatorname{tr}T_\omega \;=\; 2x$, then

$$T_\omega^{n-1} \;=\; U_{n-2}(x)T_\omega - U_{n-3}(x)I, \tag{3.13}$$

with $U_{-1} \;=\; 0$.

Proof. The eigenvalue equations at site j are

$$\begin{aligned}(\lambda+1+\omega_j)u_j^+ - \omega_j u_j^- &= u_{j-1}^+,\\ (\lambda+1+\omega_j)u_j^- - \omega_j u_j^+ &= u_{j+1}^-.\end{aligned}$$

Consequently

$$\begin{pmatrix} u_j^+ \\ u_{j+1}^- \end{pmatrix} = T_{\omega_j}(\lambda)\begin{pmatrix} u_{j-1}^+ \\ u_j^- \end{pmatrix}.$$

Periodic boundary conditions require the product monodromy to have eigenvalue one. Its determinant is one, so this is equivalent to trace two. To identify the polynomial, retain temporarily arbitrary rates $r_0, \dots, r_{n-1}$, denote the resulting generator by $L_n(r_0, \dots, r_{n-1})$, and put $b_j = \lambda + 1 + r_j$. Let E_j^+ and E_j^- denote the two equation residuals, with their right-hand sides moved to the left. For $b_j \neq 0$, replace each row pair by

$$\begin{pmatrix} F_j^+ \\ F_j^- \end{pmatrix} = \begin{pmatrix} b_j^{-1} & 0 \\ -r_j b_j^{-1} & -1 \end{pmatrix}\begin{pmatrix} E_j^+ \\ E_j^- \end{pmatrix}.$$

This block row operation has determinant $-b_j^{-1}$. If $y_j = (u_j^+, u_{j+1}^-)^\mathsf{T}$, the transformed equations are exactly

$$y_j - T_{r_j}(\lambda) y_{j-1} = 0, \qquad j \in \mathbb{Z}/n\mathbb{Z}. \tag{3.14}$$

Let A be the original coefficient matrix, F the matrix after these row operations, and C the matrix of (3.14) with columns ordered as $y_0, y_1, \dots, y_{n-1}$. The column reordering cyclically sends $(u_0^-, u_1^-, \dots, u_{n-1}^-)$ to $(u_1^-, \dots, u_{n-1}^-, u_0^-)$ and fixes the plus variables, so its sign is $(-1)^{n-1}$. Therefore

$$\det F = \frac{(-1)^n}{\prod_j b_j} \det A, \qquad \det C = (-1)^{n-1} \det F.$$

Successively eliminating $y_0, \dots, y_{n-2}$ in (3.14), using its identity diagonal blocks, leaves $(I - M) y_{n-1} = 0$, where $M = T_{r_{n-1}} \cdots T_{r_0}$; hence $\det C = \det(I - M)$. Combining the three determinant identities gives

$$\det A = -\prod_j b_j \det(I - M) = -\prod_j b_j \det(M - I).$$

Since $\det M = 1$, one has $-\det(M - I) = \operatorname{tr} M - 2$. Consequently

$$\det(\lambda I - L_n(r_0, \dots, r_{n-1})) = \left(\prod_{j=0}^{n-1} b_j\right)\left[\operatorname{tr}(T_{r_{n-1}} \cdots T_{r_0}) - 2\right]. \tag{3.15}$$

This is an identity of rational functions where all $b_j \neq 0$. The displayed product clears every denominator on the right, giving a polynomial, so the identity extends to every λ. Taking $r_0 = \omega_0$ and all other rates equal to ω, and using cyclicity of the trace, gives (3.11), including algebraic multiplicities. Equation (3.13) is the Cayley–Hamilton identity for a unimodular 2×2 matrix. ■

4. Parity factorization and finite-ring exceptional points

4.1. Reflection and parity polynomials

Definition 4.1 (Kinetic reflection)**.** Define the involution

$$(Ru)^{+}_{j} \;=\; u^{-}_{-j}, \qquad (Ru)^{-}_{j} \;=\; u^{+}_{-j}.$$

Write $\mathcal{H}_{\pm} \;=\; \ker(R \mp I)$.

Lemma 4.2 *(Protected sector)***.** The generator commutes with R, the defect vector satisfies $Rd \;=\; -d$, and

$$L|_{\mathcal{H}_+} \;=\; L^{(0)}|_{\mathcal{H}_+}.$$

Thus the defect changes only the reflection-odd sector.

Proof. Reflection reverses position and orientation simultaneously, leaving the directed hopping rules and the rate field ω_j invariant. At the fixed site zero it exchanges the two entries of d, hence $Rd \;=\; -d$. Lemma 2.2 then vanishes on $\mathcal{H}_+$. ■

We call $\mathcal{H}_+$ the protected sector and $\mathcal{H}_-$ the affected sector. These terms refer only to the response to the defect: both sectors retain their full homogeneous spectral dependence on ω.

Definition 4.3 (Parity polynomials)**.** Put $m \;=\; \lfloor (n-1)/2 \rfloor$ and let ε_n be one for even n and zero for odd n. Define

$$H_{n,+}(\lambda) \;=\; \lambda(\lambda+2)^{\varepsilon_n} \prod_{k=1}^{m} D_k(\lambda), \tag{4.1}$$

$$H^{(0)}_{n,-}(\lambda) \;=\; (\lambda+2\omega)(\lambda+2+2\omega)^{\varepsilon_n} \prod_{k=1}^{m} D_k(\lambda), \tag{4.2}$$

$$G_n(\lambda) \;=\; \frac{1}{n}\left[\frac{1}{\lambda+2\omega} + \frac{\varepsilon_n}{\lambda+2+2\omega} + 2\sum_{k=1}^{m} \frac{\lambda+1-c_k}{D_k(\lambda)}\right], \tag{4.3}$$

$$Q_{n,-}(\lambda) \;=\; H^{(0)}_{n,-}(\lambda)\big(1+2\delta G_n(\lambda)\big). \tag{4.4}$$

Each displayed denominator in (4.4) divides $H^{(0)}_{n,-}$, so $Q_{n,-}$ is a monic polynomial of degree n. An entirely polynomial version is

$$Q_{n,-} \;=\; H^{(0)}_{n,-} + \frac{2\delta}{n}\left[\frac{H^{(0)}_{n,-}}{\lambda+2\omega} + \varepsilon_n \frac{H^{(0)}_{n,-}}{\lambda+2+2\omega} + 2\sum_{k=1}^{m}(\lambda+1-c_k)\frac{H^{(0)}_{n,-}}{D_k}\right]. \tag{4.5}$$

Theorem 4.4 *(Exact parity factorization)***.** The characteristic polynomial factors as

$$P_n(\lambda;\omega,\omega_0) \;=\; H_{n,+}(\lambda) Q_{n,-}(\lambda). \tag{4.6}$$

The first factor is the characteristic polynomial on $\mathcal{H}_+$; the second is the characteristic polynomial on $\mathcal{H}_-$.

Proof. The modes q_k and $-q_k$ form a four-dimensional Fourier subspace, which splits into two isomorphic two-dimensional R-eigenspaces. Each has characteristic polynomial D_k. At $q = 0$, the vectors $(1, 1)$ and $(1, -1)$ have parities $+$ and $-$ and eigenvalues 0 and -2ω. When n is even, the $q = \pi$ density and polarization vectors have parities $+$ and $-$ and eigenvalues -2 and $-2 - 2\omega$. This gives (4.1)–(4.2). Lemma 4.2 and the rank-one determinant lemma restricted to $\mathcal{H}_-$ give (4.4). Multiplication proves (4.6). ■

4.2. Jordan structure and the exceptional locus

Definition 4.5 (Exceptional point)**.** For an eigenvalue μ, let $m_{\mathrm{alg}}(\mu)$ be its algebraic multiplicity and $m_{\mathrm{geo}}(\mu) = \dim\ker(\mu I - L)$ its geometric multiplicity. We call (ω, ω_0, μ) an exceptional point when $m_{\mathrm{alg}}(\mu) > m_{\mathrm{geo}}(\mu)$. Write $J_r(\mu)$ for a Jordan block of size r at μ.

Lemma 4.6 *(No inter-wave coincidences away from the flat band)***.** If $c_k \neq c_\ell$ and $D_k(\mu) = D_\ell(\mu) = 0$, then $\omega = 1$ and $\mu = -2$.

Proof. Subtracting the two equations gives $2a(c_k - c_\ell) = 0$, hence $a = 0$. Substitution into (3.3) gives $1 - \omega^2 = 0$. Since $\omega > 0$, this means $\omega = 1$ and then $\mu = -1 - \omega = -2$. ■

Lemma 4.7 *(Scalar resolvent criterion for cyclicity)***.** Let $A \in \mathbb{C}^{N\times N}$ and $v \in \mathbb{C}^N$. Suppose the reduced denominator of the scalar rational function

$$v^{\mathsf{T}}(\lambda I - A)^{-1} v$$

is the characteristic polynomial $\chi_A(\lambda)$. Then v is a cyclic vector for A. Moreover, for every $\eta \in \mathbb{C}$, the same vector is cyclic for $A - \eta v v^{\mathsf{T}}$.

Proof. Let $m_{A,v}$ be the monic polynomial of least degree satisfying $m_{A,v}(A)v = 0$. For $\lambda \notin \operatorname{spec} A$,

$$m_{A,v}(\lambda)(\lambda I - A)^{-1} v = \frac{m_{A,v}(\lambda) I - m_{A,v}(A)}{\lambda I - A}\, v. \tag{4.7}$$

The matrix quotient on the right is a matrix polynomial in λ: for each monomial it is the finite divided-difference identity

$$\frac{\lambda^j I - A^j}{\lambda I - A} = \sum_{r=0}^{j-1} \lambda^{j-1-r} A^r.$$

Thus the reduced denominator of every scalar component of $(\lambda I - A)^{-1} v$, and in particular of the displayed scalar resolvent, divides $m_{A,v}$. By hypothesis, χ_A therefore divides $m_{A,v}$. Conversely, Cayley–Hamilton gives $\chi_A(A)v = 0$. Write $\chi_A = q\, m_{A,v} + r$ with $\deg r < \deg m_{A,v}$. Then $r(A)v = 0$, so the defining minimality of $m_{A,v}$ forces $r = 0$. Hence $m_{A,v}$ divides χ_A, and therefore $m_{A,v} = \chi_A$ and has degree N, which is equivalent to $v, Av, \ldots, A^{N-1}v$ spanning $\mathbb{C}^N$.

Put $B = A - \eta v v^{\mathsf{T}}$. Induction on j gives

$$B^j v = A^j v + \sum_{r=0}^{j-1} c_{j,r} A^r v$$

for suitable scalars $c_{j,r}$. The change from the A-Krylov list to the B-Krylov list is therefore triangular with diagonal entries one. The two lists have the same span, so v is cyclic for B. ■

Theorem 4.8 *(Jordan classification for $\omega \neq 1$)***.** Assume $\omega \neq 1$.

1. In the protected sector, the endpoint factors in (4.1) give one-dimensional blocks. For each $1 \leq k \leq m$, a root of D_k gives one J_1, except that

$$\omega = |s_k|, \qquad \mu = c_k - 1 - \omega. \tag{4.8}$$

   gives one $J_2(\mu)$.

2. In the affected sector, if

$$Q_{n,-}(\mu) = Q'_{n,-}(\mu) = \cdots = Q^{(r-1)}_{n,-}(\mu) = 0, \qquad Q^{(r)}_{n,-}(\mu) \neq 0, \tag{4.9}$$

   then the complete affected-sector contribution at μ is one block $J_r(\mu)$.

3. The full Jordan form at μ is the direct sum of the blocks in items 1 and 2. In particular, a crossing between the two parity sectors is semisimple unless one of its sector blocks has size greater than one.

Consequently, (4.8) together with (4.9) is an exact finite algebraic criterion for every exceptional point when $\omega \neq 1$. The order r in (4.9) classifies exceptional points of arbitrary order.

Proof. The discriminant of D_k is $4(\omega^2 - s_k^2)$. When the discriminant vanishes, its Fourier block is not scalar and hence is a single J_2; otherwise each root is simple. Lemma 4.6 excludes coincidences between distinct c_k values.

It remains to show that the affected restriction is cyclic. In each two-dimensional homogeneous parity block, the spectral weight of the defect vector has numerator $2(\lambda + 1 - c_k)$. This numerator and D_k have no common root for an interior wave number: a common root would give $\lambda + 1 = c_k$, after which $D_k = s_k^2 \neq 0$. The endpoint weights in (4.3) are also nonzero. At a double root of an interior D_k, the numerator remains nonzero, so the scalar resolvent retains a pole of order two. Regard the endpoint factors as the odd-parity roots of D_0 and, when n is even, $D_{n/2}$. Lemma 4.6 then makes their root sets disjoint from those of every interior factor and from one another when $\omega \neq 1$. Consequently no pole, including a repeated pole, cancels when the block fractions are put over their common denominator. The scalar resolvent $d^{\mathsf{T}}(\lambda I - L^{(0)}|_{\mathcal{H}_-})^{-1} d = 2G_n(\lambda)$ therefore has reduced denominator $H^{(0)}_{n,-}$, the characteristic polynomial of the homogeneous odd restriction. Lemma 4.7 makes d cyclic first for that restriction and then for its update by $-\delta d d^{\mathsf{T}}$. Hence the minimal and characteristic polynomials of the affected restriction both equal $Q_{n,-}$. A cyclic matrix has one Jordan block for each eigenvalue, of size equal to the root order. The direct-sum assertion follows from the invariant parity splitting. ∎

Theorem 4.9 *(Flat-band classification at $\omega = 1$).* Let $\omega = 1$ and put $\mu = -2$. Its Jordan contribution is given by the following table, where $J_r(\mu)^{\oplus a}$ denotes the direct sum of a copies.

Parameters	Contribution at μ
$\omega_0 = 1$ and $4 \nmid n$	$J_1(\mu)^{\oplus n}$
$\omega_0 = 1$ and $4 \mid n$	$J_2(\mu)^{\oplus 2} \oplus J_1(\mu)^{\oplus(n-2)}$
$\omega_0 \neq 1$ and $n \equiv 1, 2 \pmod 4$	$J_1(\mu)^{\oplus(n-1)}$
$\omega_0 \neq 1$ and $n \equiv 0 \pmod 4$	$J_2(\mu) \oplus J_1(\mu)^{\oplus(n-2)}$
$n \equiv 3 \pmod 4$, $\omega_0 \neq 1$, and $\omega_0 \neq (n-1)/(n+1)$	$J_2(\mu) \oplus J_1(\mu)^{\oplus(n-2)}$
$n \equiv 3 \pmod 4$ and $\omega_0 = (n-1)/(n+1)$	$J_3(\mu) \oplus J_1(\mu)^{\oplus(n-2)}$

For eigenvalues $\lambda \neq -2$, every protected-sector root is a one-dimensional block. Remove the exact power of $(\lambda + 2)$ specified above from $Q_{n,-}$. Every root of the remaining affected polynomial contributes one Jordan block, so the derivative test (4.9) applied to that reduced polynomial classifies every remaining affected block at $\omega = 1$. The full Jordan form is the direct sum of the protected and affected blocks.

Proof. Algebraic multiplicity at the flat point. Write $t = \lambda + 2$. At $\omega = 1$,

$$D_k = t(t - 2c_k), \qquad P_n = t^{n-1} F_n(t),$$

where, for $\delta = \omega_0 - 1$,

$$F_n(t) = 2t\big(T_n(t/2) - 1\big) + \delta\left[2\big(T_n(t/2) - 1\big) + (t-2)U_{n-1}(t/2)\right]. \tag{4.10}$$

Put $C_n = \cos(n\pi/2)$ and $S_n = \sin(n\pi/2)$. The values $T_n(0) = C_n$, $U_{n-1}(0) = S_n$, together with

$$T_n'(0) = nS_n, \qquad U_{n-1}'(0) = -nC_n,$$

give directly from (4.10)

$$F_n(0) = 2\delta(C_n - 1 - S_n), \tag{4.11}$$

$$F_n'(0) = 2(C_n - 1) + \delta\big((n+1)S_n + nC_n\big). \tag{4.12}$$

If $\delta = 0$ and $4 \nmid n$, the derivative in (4.12) is nonzero. If $\delta = 0$ and $4 \mid n$, the Chebyshev differential equation at zero gives $T_n''(0) = -n^2$, and hence

$$F_n(t) = -\frac{n^2}{4}t^3 + O(t^5).$$

For $\delta \neq 0$, equations (4.11)–(4.12) give

$n \pmod 4$	first nonzero datum	$\operatorname{ord}_{t=0} F_n$
$1, 2$	$F_n(0) = -4\delta$	0
0	$F_n'(0) = n\delta$	1
3	$F_n'(0) = -2 - (n+1)\delta$	1

in the last row unless $\delta = -2/(n+1)$. For $n \equiv 3 \pmod 4$ one has $U_{n-1}''(0) = n^2 - 1$, and direct differentiation of (4.10) at that tuned value gives

$$F_n''(0) = -2n + \frac{U_{n-1}''(0)}{n+1} = -(n+1) \neq 0. \tag{4.13}$$

Thus $\operatorname{ord}_0 F_n$ is respectively 1 or 3 when $\delta = 0$ according as $4 \nmid n$ or $4 \mid n$, and is respectively 0, 1, 1, or 2 in the four defective cases displayed above, with the final value belonging to the tuned $n \equiv 3 \pmod 4$ case. Since $P_n = t^{n-1}F_n$, this proves the stated algebraic multiplicities, including the tuned multiplicity $n+1$.

Geometric multiplicity. The first bulk equation at each $j \neq 0$ and the second bulk equation at $j = n-1$ give

$$u_j^- = -u_{j-1}^+ \quad (1 \le j \le n-1), \qquad u_0^- = -u_{n-1}^+.$$

The remaining bulk equations repeat these relations, so all minus components are determined by the n plus components. Put $\delta = \omega_0 - 1$. The two defect equations are

$$\delta u_0^+ - \omega_0 u_0^- = u_{n-1}^+, \qquad \delta u_0^- - \omega_0 u_0^+ = u_1^- = -u_0^+.$$

Substituting $u_0^- = -u_{n-1}^+$ shows that these equations are, respectively,

$$\delta(u_0^+ + u_{n-1}^+) = 0, \qquad -\delta(u_0^+ + u_{n-1}^+) = 0.$$

Thus there is no constraint on the n plus components when $\omega_0 = 1$, whereas $u_{n-1}^+ = -u_0^+$ is the single constraint when $\omega_0 \neq 1$. The geometric multiplicity is therefore n in the homogeneous case and $n-1$ otherwise. Fourier decomposition shows that the two excess algebraic dimensions when $4 \mid n$ and $\delta = 0$ are the two $q = \pm\pi/2$ blocks, each a J_2. For $\delta \neq 0$, the parity factorization places the $4 \mid n$ excess in one protected J_2.

Size of the nontrivial defective block. It remains to distinguish one J_3 from two J_2 blocks in the tuned $n \equiv 3 \pmod 4$ case. Put $N_0 = L_n(1,1) + 2I$ and $N = L_n(1,\omega_0) + 2I = N_0 - \delta d d^{\mathsf T}$. Because $4 \nmid n$, zero is a semisimple eigenvalue of N_0 with eigenspace $E = \ker N_0$ of dimension n. Let Π be its spectral projection. Write

$$\det(tI - N_0) = t^n h(t), \qquad h(0) \neq 0.$$

Semisimplicity gives $(tI - N_0)^{-1} = t^{-1}\Pi + O(1)$. The rank-one determinant formula then shows that the coefficient of t^{n-1} in $\det(tI - N)$ is $\delta h(0) d^{\mathsf T}\Pi d$. The multiplicity computation above shows that this coefficient vanishes for $n \equiv 3 \pmod 4$, and therefore

$$d^{\mathsf T}\Pi d = 0. \tag{4.14}$$

The bulk relations derived above show directly that the eigenspace of N is $K = E \cap \ker d^{\mathsf T}$, which has dimension $n-1$. If $v \in K \cap \operatorname{ran} N$, write $v = Nu$ and apply Π:

$$v = \Pi N u = -\delta\, \Pi d\, (d^{\mathsf T} u). \tag{4.15}$$

Thus $K \cap \operatorname{ran} N$ has dimension at most one. For a finite matrix, every Jordan block of size at least two contributes its leading eigenvector to $\ker N \cap \operatorname{ran} N$, these vectors are independent, and they span that intersection. The algebraic multiplicity exceeds the geometric multiplicity here, so the intersection has dimension one. Hence exactly one Jordan block at zero for N is nontrivial. It is a J_2 when the algebraic multiplicity is n, and the additional algebraic order at $\delta = -2/(n+1)$ makes that same block a J_3. This proves all six cases and excludes the alternative of two J_2 blocks in the tuned case.

Affected roots away from the flat band. Put $A_-^{(0)} = L^{(0)}|_{\mathcal{H}_-}$ and consider an affected eigenvalue $\lambda \neq -2$. If $\delta \neq 0$, it cannot equal an eigenvalue away from -2 of the homogeneous affected restriction $A_-^{(0)}$: the corresponding residue in (4.3) is nonzero, so $Q_{n,-}$ does not vanish at that pole. Hence $\lambda I - A_-^{(0)}$ is invertible. The rank-one eigenvalue equation makes its eigenspace the one-dimensional span of $(\lambda I - A_-^{(0)})^{-1} d$. Its root order therefore gives the size of its unique Jordan block. If $\delta = 0$, all roots away from -2 are simple by Lemma 4.6. This proves the final classification without asserting cyclicity at the removed flat primary component. ■

Corollary 4.10 *(Finite-ring algebraic exceptional locus)***.** For fixed n and $\omega \neq 1$, the affected exceptional parameters are exactly those satisfying

$$\operatorname*{Res}_{\lambda}\bigl(Q_{n,-}, \partial_\lambda Q_{n,-}\bigr) = 0,$$

with $Q_{n,-}$ given by (4.5). At $\omega = 1$, put

$$r_- = \begin{cases} (n+1)/2, & \omega_0 = 1,\ n \text{ odd}, \\ n/2, & \omega_0 = 1,\ n \equiv 2 \pmod 4, \\ n/2+1, & \omega_0 = 1,\ n \equiv 0 \pmod 4, \\ (n-1)/2, & \omega_0 \neq 1,\ n \equiv 1 \pmod 4, \\ n/2-1, & \omega_0 \neq 1,\ n \text{ even}, \\ (n+1)/2, & \omega_0 \neq 1,\ n \equiv 3 \pmod 4,\ \omega_0 \neq (n-1)/(n+1), \\ (n+3)/2, & n \equiv 3 \pmod 4,\ \omega_0 = (n-1)/(n+1), \end{cases} \tag{4.16}$$

and define

$$\widehat{Q}_{n,-}(\lambda) = \frac{Q_{n,-}(\lambda)}{(\lambda+2)^{r_-}}.$$

If $n = 3$ and $\omega_0 = 1/2$, then $r_- = 3$ and $\widehat{Q}_{3,-} = 1$, so there are no affected eigenvalues away from $\lambda = -2$. On every other parameter stratum in (4.16), the affected exceptional parameters away from $\lambda = -2$ are exactly those satisfying

$$\operatorname*{Res}_{\lambda}\bigl(\widehat{Q}_{n,-}, \partial_\lambda \widehat{Q}_{n,-}\bigr) = 0.$$

The protected exceptional parameters (4.8) and the flat-band cases of Theorem 4.9 are listed separately; they are not asserted to arise from these affected-sector resultants. Root orders and Theorems 4.8–4.9 distinguish exceptional points from semisimple parity crossings without a numerical tolerance. For the general affected sector, this is an implicit algebraic classification: the resultant is an exact finite polynomial equation for the exceptional locus, and the root-order tests determine the Jordan block, but no parametric solution of that resultant equation is asserted.

Proof. For a nonconstant polynomial F, the resultant $\mathrm{Res}_\lambda(F, F')$ vanishes exactly when F has a repeated root. For $\omega \neq 1$, the polynomial $Q_{n,-}$ is monic of degree n, so Theorem 4.8 proves the first assertion. At $\omega = 1$, it remains to verify the exponent in (4.16). Definition 4.3 gives

$$\operatorname{ord}_{\lambda=-2} H_{n,+} = \begin{cases} (n-1)/2, & n \text{ odd}, \\ n/2, & n \equiv 2 \pmod 4, \\ n/2+1, & n \equiv 0 \pmod 4. \end{cases}$$

Subtracting these protected orders from the total flat algebraic multiplicities in Theorem 4.9 gives every line of (4.16). Thus $\widehat{Q}_{n,-}$ removes the exact affected flat primary factor on each stratum. At the tuned $n = 3$ point, $Q_{3,-}$ is monic of degree three and its exact flat order is three, so $\widehat{Q}_{3,-} = 1$ and the affected spectrum away from $\lambda = -2$ is empty. On every other stratum, $\widehat{Q}_{n,-}$ is nonconstant, and the resultant criterion applies. Theorems 4.8–4.9 then identify the Jordan block attached to every repeated root away from -2 and separately account for the protected and flat contributions. ■

5. Defect-localized modes

5.1. Bulk spectrum and Green function

Definition 5.1 (Bulk spectral set)**.** For the infinite translation-invariant chain, define

$$\Sigma_\omega = \left\{ c - 1 - \omega \pm \sqrt{\omega^2 - (1-c^2)} \; : \; c \in [-1,1] \right\}.$$

When $a \neq 0$, the identity $D_k = 2a(x - c_k)$, with c_k replaced by a continuous parameter $c \in [-1, 1]$, shows that

$$\lambda \in \Sigma_\omega \quad \Longleftrightarrow \quad x(\lambda) \in [-1,1]. \tag{5.1}$$

For $\lambda \notin \Sigma_\omega$ with $a \neq 0$, retain a and x from (3.1)–(3.2), and choose

$$s(\lambda) = \sqrt{x(\lambda)^2 - 1}$$

as the standard branch on $x \in \mathbb{C} \setminus [-1, 1]$ satisfying $s/x \to 1$ as $|x| \to \infty$. Equivalently, it is the branch satisfying $s(\lambda) \sim x(\lambda)$ as $|\lambda| \to \infty$. Put $z = x - s$, so $|z| < 1$.

Let $L_\infty^{(0)}$ act on $\ell^2(\mathbb{Z}; \mathbb{C}^2)$ by (2.1)–(2.2) with every local rate equal to ω, and let $L_\infty = L_\infty^{(0)} - \delta d d^{\mathsf{T}}$, where now d is the polarization vector at the origin. These operators are bounded because they are finite linear combinations of bilateral shifts and bounded multiplication operators. Infinite-line eigenvalues below always mean ℓ^2 eigenvalues of L_∞.

Lemma 5.2 *(Bulk spectrum and cyclic Green identities)***.** The homogeneous infinite-chain spectrum is exactly

$$\operatorname{spec} L_\infty^{(0)} = \Sigma_\omega.$$

For $x \in \mathbb{C} \setminus [-1,1]$, with $s = \sqrt{x^2-1}$ and $z = x - s$ on the branch of Definition 5.1,

$$\frac{1}{2\pi}\int_0^{2\pi} \frac{dq}{x-\cos q} \;=\; \frac{1}{s}, \tag{5.2}$$

$$\frac{1}{n}\sum_{k=0}^{n-1} \frac{1}{x-\cos q_k} \;=\; \frac{1}{s}\frac{1+z^n}{1-z^n}. \tag{5.3}$$

The right side of (5.3) is invariant under the simultaneous branch change $(s,z) \mapsto (-s, z^{-1})$ and therefore extends meromorphically to every $x \in \mathbb{C}$ away from the finite set $\{\cos q_k : 0 \le k < n\}$, where it equals the finite sum on the left.

Proof. The unitary Fourier transform identifies $L_\infty^{(0)}$ with multiplication on $L^2([0,2\pi];\mathbb{C}^2)$ by the continuous matrix symbol

$$W(q) \;=\; \begin{pmatrix} e^{\mathrm{i}q} - 1 - \omega & \omega \\ \omega & e^{-\mathrm{i}q} - 1 - \omega \end{pmatrix}.$$

Its two pointwise eigenvalues are

$$\cos q - 1 - \omega \pm \sqrt{\omega^2 - \sin^2 q},$$

whose union is Σ_ω. If $\lambda \notin \Sigma_\omega$, then $\lambda I - W(q)$ is invertible for every q and its inverse is a continuous matrix function on the compact circle, hence is uniformly bounded and defines $(\lambda - L_\infty^{(0)})^{-1}$. Conversely, if λ is a pointwise eigenvalue at q_0, choose a corresponding nonzero vector and normalized scalar functions supported in shrinking neighborhoods of q_0. The resulting Fourier-space vectors form an approximate eigenvector sequence by continuity of W, so $\lambda \in \operatorname{spec} L_\infty^{(0)}$. This proves the spectral identity.

For (5.2), put $\zeta = e^{\mathrm{i}q}$. Since $z + z^{-1} = 2x$ and $|z| < 1$, the two roots of the resulting denominator are z and z^{-1}, and only z lies inside the unit circle. The residue theorem gives

$$\frac{1}{2\pi}\oint_{|\zeta|=1} \frac{2\mathrm{i}\, d\zeta}{(\zeta - z)(\zeta - z^{-1})} \;=\; \frac{2}{z^{-1}-z} \;=\; \frac{1}{s}.$$

Also, the absolutely convergent Poisson-kernel expansion is

$$\frac{1}{x-\cos q} \;=\; \frac{1}{s}\left(1 + 2\sum_{r=1}^{\infty} z^r \cos(rq)\right).$$

Averaging over $q_k = 2\pi k/n$ removes every term for which $n \nmid r$. Summing the remaining geometric series proves (5.3). Finally, the displayed branch change fixes its right side. It consequently defines a single-valued meromorphic function, which agrees with the finite rational sum on $\mathbb{C} \setminus [-1,1]$ and hence everywhere away from their poles. ∎

Proposition 5.3 *(Embedded point spectrum and singular flat band)***.** Let $\lambda \in \Sigma_\omega$.

1. If $a = \lambda + 1 + \omega \neq 0$, then $\ker_{\ell^2}(L_\infty - \lambda I) = \{0\}$.

2. If $a = 0$, then necessarily $\omega = 1$ and $\lambda = -2$. For every $\omega_0 \geq 0$, the embedded flat-band eigenspace is exactly

$$\ker_{\ell^2}(L_\infty + 2I) = \left\{ u : \begin{array}{l} u_j^+ = p_j, \quad u_j^- = -p_{j-1}, \quad p \in \ell^2(\mathbb{Z}), \\ (\omega_0 - 1)(p_0 + p_{-1}) = 0 \end{array} \right\}. \tag{5·4}$$

This eigenspace is infinite dimensional. Its compactly supported vectors are precisely those obtained from finitely supported p.

Thus the singular flat band is the only embedded ℓ^2 point spectrum; all other infinite-line localized eigenvalues lie outside Σ_ω.

Proof. Suppose first that $a \neq 0$. By (5.1), $x \in [-1, 1]$. For an eigenvector u, put

$$y_j = \begin{pmatrix} u_j^+ \\ u_{j+1}^- \end{pmatrix}.$$

Away from the defect, the transfer relation in the proof of Theorem 3.4 gives

$$y_j = T_\omega(\lambda) y_{j-1}, \qquad j \neq 0. \tag{5·5}$$

The matrix T_ω is unimodular and has trace $2x$. If $x \in (-1, 1)$, its two eigenvalues are distinct and lie on the unit circle, so the families T_ω^m and T_ω^{-m} are uniformly bounded. Consequently, neither a forward nor a backward orbit can tend to zero unless its initial vector is zero. At $x = \varepsilon \in \{-1, 1\}$, Cayley–Hamilton gives

$$T_\omega = \varepsilon I + N, \qquad N^2 = 0, \qquad T_\omega^m = \varepsilon^m \big(I + m\varepsilon^{-1} N\big).$$

If $Nv = 0$, then $\|T_\omega^m v\| = \|v\|$; if $Nv \neq 0$, then

$$\|T_\omega^m v\| \geq m\|Nv\| - \|v\| \longrightarrow \infty.$$

The identity $T_\omega^{-m} = \varepsilon^{-m}(I - m\varepsilon^{-1}N)$ gives the same alternatives for negative powers. Hence an orbit in either direction tends to zero only when $v = 0$.

Because $u \in \ell^2$, one has $y_j \to 0$ as $j \to \pm\infty$. Applying the preceding orbit alternatives to the right half-line in (5.5) gives $y_0 = 0$, and applying them to the left half-line gives $y_{-1} = 0$. These two identities and the recurrence force every component of u to vanish. This proves the first item.

Now suppose $a = 0$. Then $\lambda = -1 - \omega$ and the bulk determinant is $D(q) = 1 - \omega^2$, independent of q. Membership in Σ_ω therefore forces $\omega = 1$ and $\lambda = -2$. Set $p_j = u_j^+$. At every bulk site the eigenvalue equations reduce to

$$u_j^- = -p_{j-1}, \qquad u_{j+1}^- = -p_j.$$

The equations adjacent to the defect supply the same relation across both interfaces, so $u_j^- = -p_{j-1}$ for every $j \in \mathbb{Z}$. After this substitution, the two equations at the defect are both equivalent to

$$(\omega_0 - 1)(p_0 + p_{-1}) = 0.$$

Conversely, any $p \in \ell^2(\mathbb{Z})$ satisfying this scalar condition gives an ℓ^2 eigenvector by direct substitution. This proves (5.4). The condition cuts out either all of $\ell^2(\mathbb{Z})$ or one codimension-one subspace, so the eigenspace is infinite dimensional; the support assertion follows from the displayed shift relation. ■

5.2. Infinite-line defect modes

A localized mode whose support is infinite is called an *infinite-support localized mode.* In the decay-root parametrization below this is the range $0 < |z| < 1$; the compact case $z = 0$ is treated separately.

Theorem 5.4 *(Infinite-line localization criterion outside the bulk spectrum)***.** Let $\delta \neq 0$, $\lambda \notin \Sigma_\omega$, and $a \neq 0$. Then λ is an eigenvalue of L_∞ admitting an exponentially localized ℓ^2 eigenvector if and only if

$$1 + 2\delta g_\infty(\lambda) = 0, \tag{5.6}$$

where

$$g_\infty(\lambda) = \frac{1}{2a}\left(1 + \frac{\lambda + 1 - x}{s}\right) = \frac{z(\lambda + 1 - z)}{a(1 - z^2)}. \tag{5.7}$$

The localization length is

$$\xi(\lambda) = \frac{-1}{\log |z|}.$$

Equivalently, with $a = \lambda + 1 + \omega$, the pair (a, z) satisfies

$$a^2 + 1 - \omega^2 - a(z + z^{-1}) = 0, \tag{5.8}$$

$$a(1 - z^2) + 2\delta z(a - \omega - z) = 0, \qquad 0 < |z| < 1. \tag{5.9}$$

Proof. The polarization entry of the bulk resolvent at the defect site is

$$\begin{aligned} g_\infty(\lambda) &= \frac{1}{2\pi}\int_0^{2\pi} \frac{\lambda + 1 - \cos q}{a^2 + 1 - \omega^2 - 2a\cos q}\, dq \\ &= \frac{1}{2a}\left[1 + (\lambda + 1 - x)\frac{1}{2\pi}\int_0^{2\pi} \frac{dq}{x - \cos q}\right] \\ &= \frac{1}{2a}\left(1 + \frac{\lambda + 1 - x}{s}\right). \end{aligned}$$

Here the last equality is (5.2). If $(\lambda - L_\infty)u = 0$, then

$$u = -\delta(\lambda - L_\infty^{(0)})^{-1} d\,(d^{\mathsf{T}} u).$$

The scalar $d^{\mathsf{T}} u$ cannot vanish unless $u = 0$. Applying d^{T} therefore gives (5.6). Conversely, when that scalar equation holds, the displayed resolvent vector is a nonzero ℓ^2 eigenvector. Since $x = (z + z^{-1})/2$ and $s = (z^{-1} - z)/2$, elementary simplification gives the second expression in (5.7). The Fourier coefficient of $(x - \cos q)^{-1}$ is $z^{|j|}/s$; consequently, on the half-line $j \geq 0$, the second component of $(\lambda - L_\infty^{(0)})^{-1} d$ is

$$-\frac{\lambda + 1 - z}{2as}\, z^j.$$

The secular equation and $\delta \neq 0$ give $g_\infty(\lambda) = -1/(2\delta) \neq 0$. The second identity in (5.7), together with $a \neq 0$, $z \neq 0$, and $|z| < 1$, therefore gives $\lambda + 1 - z \neq 0$. This tail does not cancel, so its exact exponential rate is $|z|$ and the localization length is $-1/\log|z|$. Clearing the denominator in (5.6) gives (5.9); the bulk dispersion relation gives (5.8). ∎

Corollary 5.5 *(Algebraic candidates for infinite-support localized modes)***.** Under the hypotheses of Theorem 5.4, every admissible z is a root of

$$\begin{aligned} 0 \;=\; & 2\delta z^3 + (2\delta\omega + 1 - \omega^2)z^2 \\ & + (4\delta^2\omega + 4\delta\omega^2 - 2\delta)z + (2\delta\omega + \omega^2 - 1), \end{aligned} \tag{5.10}$$

and, when the denominator is nonzero,

$$a \;=\; \frac{2\delta z(\omega + z)}{1 - z^2 + 2\delta z}, \qquad \lambda \;=\; a - 1 - \omega. \tag{5.11}$$

The zero-denominator branch in (5.11) occurs precisely when

$$0 \;<\; \omega \;<\; 1, \qquad \omega_0 \;=\; \frac{\omega^2 + 1}{2\omega}, \qquad z \;=\; -\omega. \tag{5.12}$$

At these parameters there are two admissible values

$$a_\pm \;=\; \frac{-\omega^2 - 1 \pm \sqrt{5\omega^4 - 2\omega^2 + 1}}{2\omega}, \qquad \lambda_\pm \;=\; a_\pm - 1 - \omega, \tag{5.13}$$

and both give infinite-support localized modes with decay root $z = -\omega$. A root of (5.10) is accepted only when both (5.8) and (5.9) hold; this removes roots introduced by elimination or by clearing denominators.

Proof. After multiplication of the dispersion relation by z, the elimination factorization is

$$\mathrm{Res}_a\big(za^2 + z(1-\omega^2) - a(z^2+1), a(1 - z^2 + 2\delta z) - 2\delta z(\omega + z)\big) \;=\; z(z^2 - 1)\mathcal{C}_{\omega,\delta}(z),$$

where $\mathcal{C}_{\omega,\delta}$ is the cubic polynomial displayed in (5.10). The admissible infinite-support range $0 < |z| < 1$ excludes the three prefactors: $z = \pm 1$ are bulk-edge points, while $z = 0$ is evaluated separately below. Thus eliminating a gives (5.10) on the stated range. Solving (5.9) for a gives (5.11).

Put $C = 1 - z^2 + 2\delta z$. If $C = 0$, the matching equation reduces to $2\delta z(\omega + z) = 0$. Since $\delta \neq 0$ and $z \neq 0$, this forces $z = -\omega$, and then $C = 0$ is equivalent to $\delta = (1 - \omega^2)/(2\omega)$. The condition $|z| < 1$ is exactly $0 < \omega < 1$. Substitution in the dispersion equation gives

$$a^2 + (\omega + \omega^{-1})a + 1 - \omega^2 \;=\; 0,$$

whose two roots are the values in (5.13). Conversely, each root satisfies both (5.8) and (5.9). Moreover, the quadratic has nonzero constant term $1 - \omega^2$, so neither root has $a = 0$. Since $z = -\omega$ and $0 < \omega < 1$,

$$x \;=\; \frac{1}{2}(z + z^{-1}) \;=\; -\frac{1}{2}(\omega + \omega^{-1}) \;<\; -1.$$

Thus the corresponding λ lies outside Σ_ω, and every hypothesis of Theorem 5.4 is satisfied. Both roots therefore give the asserted infinite-support localized modes. At this tuning the cubic itself factors as

$$\frac{1-\omega^2}{\omega}\, z(z+\omega)^2,$$

so its double nonzero root accounts for the two values of a rather than for a repeated eigenvalue. ■

Proposition 5.6 *(Compact singular-transfer mode)***.** Let $\omega \neq 1$. The value $z = 0$, excluded from the infinite-support parametrization, gives

$$\lambda = -1-\omega \quad \text{as a localized eigenvalue if and only if} \quad \omega_0 = \frac{\omega^2+1}{2\omega}. \tag{5.14}$$

Its localization length is $\xi = 0$ by the finite-support convention, and its eigenvector is supported at the defect and its two neighboring sites.

Proof. At $\lambda = -1-\omega$ the Fourier denominator is the nonzero constant $1-\omega^2$, and hence

$$g_\infty(-1-\omega) = \frac{1}{2\pi}\int_0^{2\pi} \frac{-\omega-\cos q}{1-\omega^2}\, dq = \frac{\omega}{\omega^2-1}.$$

The secular equation is therefore equivalent to $\delta = (1-\omega^2)/(2\omega)$, which is exactly (5.14). In Fourier variables the associated resolvent vector is proportional to

$$\begin{pmatrix} -\omega - e^{-\mathrm{i}q} \\ \omega + e^{\mathrm{i}q} \end{pmatrix}.$$

Its inverse Fourier transform is supported at the defect and its two neighbors. This proves the compact case and also shows directly that it is the limiting root $z = 0$. ■

5.3. Return to finite rings

Proposition 5.7 *(Finite-ring convergence of a localized mode)***.** Suppose (5.6) has a simple root λ_∞ with decay root z_∞, $0 < |z_\infty| < 1$. There is a closed disk B centered at λ_∞, disjoint from Σ_ω, such that, for all sufficiently large n, the finite polynomial $Q_{n,-}$ has exactly one zero λ_n in the interior of B, counted with algebraic multiplicity, and

$$\lambda_n - \lambda_\infty = O(|z_\infty|^n).$$

If the corresponding right eigenvector is normalized in ℓ^2, define its squared site amplitude by $M_j(u) = |u_j^+|^2 + |u_j^-|^2$. This is a right-eigenvector localization diagnostic, not a probability law or stationary mass of the Markov chain; no assertion about left-eigenvector localization is made. The amplitude is bounded by $C|z_\infty|^{2\,\mathrm{dist}(j,0)}$ for a constant C independent of j and of all sufficiently large n. In the compact case (5.14), the infinite and finite eigenvalues agree exactly and the eigenvector is supported where $\mathrm{dist}(j,0) \leq 1$.

Proof. Eigenvalue convergence. Equation (5.3) applied to the bulk Green kernel gives the exact identity

$$G_n(\lambda) \;=\; \frac{1}{2a}\left[1+\frac{\lambda+1-x}{s}\,\frac{1+z^n}{1-z^n}\right]. \tag{5.15}$$

Under the simultaneous branch change $(s,z)\mapsto(-s,z^{-1})$, the last two factors in the second term both change sign. The right-hand side is therefore branch invariant and continues meromorphically as the finite rational function G_n to every point away from a finite-ring homogeneous pole. This continuation statement concerns G_n only; the infinite-volume resolvent entry g_∞ remains defined on $\mathbb{C}\setminus\Sigma_\omega$. On the closed disk chosen below, which is disjoint from Σ_ω, use the common branch of s and z. Subtracting (5.7) from (5.15) then gives

$$G_n - g_\infty \;=\; \frac{\lambda+1-x}{as}\,\frac{z^n}{1-z^n}. \tag{5.16}$$

Choose a closed disk B about λ_∞ disjoint from Σ_ω and containing no other zero of $1+2\delta g_\infty$. Shrinking it if necessary, choose

$$\rho_B \;=\; \max_{\lambda\in B}|z(\lambda)| \;<\; r \;<\; 1.$$

By Lemma 3.2, every zero of $H^{(0)}_{n,-}$ is a finite-ring homogeneous Fourier eigenvalue and therefore belongs to Σ_ω. Hence $H^{(0)}_{n,-}$ is nonzero on B, and the zeros of $1+2\delta G_n$ in B are exactly the zeros of $Q_{n,-}$ there. Equation (5.16) and its λ-derivative are bounded on B by $C_B r^n$: for the derivative, the only additional factor is $n\rho_B^{n-1} = O(r^n)$. Rouché's theorem on ∂B, followed by the simple-zero estimate at λ_∞, gives one zero λ_n and first yields $\lambda_n-\lambda_\infty = O(r^n)$. The root equation and (5.16) then give

$$|\lambda_n-\lambda_\infty| \;\le\; C|z(\lambda_n)|^n.$$

Since $z_\infty\neq 0$ and $z(\lambda_n)/z_\infty = 1+O(r^n)$, one has $|z(\lambda_n)/z_\infty|^n = \exp(O(nr^n)) = O(1)$. Hence $\lambda_n-\lambda_\infty = O(|z_\infty|^n)$.

Uniform localization. Put $\rho_\infty = |z_\infty|$, $z_n = z(\lambda_n)$, and $s_n = s(\lambda_n)$. Averaging the Poisson-kernel expansion in Lemma 5.2 against $e^{\mathrm{i}jq_k}$ gives, for the representative $j\in\{0,\dots,n-1\}$,

$$\frac{1}{n}\sum_{k=0}^{n-1}\frac{e^{\mathrm{i}jq_k}}{x(\lambda_n)-\cos q_k} \;=\; \frac{1}{s_n}\sum_{\ell\in\mathbb{Z}} z_n^{|j+\ell n|}. \tag{5.17}$$

Indeed, the Fourier coefficient on the infinite circle is $z_n^{|j|}/s_n$, and root-of-unity averaging retains exactly the indices congruent to $-j$ modulo n. Moreover, in Fourier variables,

$$(\lambda_n - W(q))^{-1}d \;=\; \frac{1}{2a_n(x(\lambda_n)-\cos q)}\begin{pmatrix}\lambda_n+1-e^{-\mathrm{i}q}\\ -\lambda_n-1+e^{\mathrm{i}q}\end{pmatrix}, \qquad a_n \;=\; \lambda_n+1+\omega.$$

Thus each component of the finite-ring resolvent vector is a bounded linear combination of the coefficients in (5.17) at j and $j\pm1$. If $h_j = \mathrm{dist}(j,0)\le n/2$, then the geometric series gives the exact majorant

$$\sum_{\ell\in\mathbb{Z}}|z_n|^{|j+\ell n|} \;=\; \frac{|z_n|^{h_j}+|z_n|^{n-h_j}}{1-|z_n|^n}.$$

Since s_n and a_n stay away from zero and $|z_n| \geq \rho_\infty/2$ for large n, the shifts by one only alter the constant. Consequently the periodized resolvent components are bounded by

$$C\frac{|z_n|^{h_j} + |z_n|^{n-h_j}}{1 - |z_n|^n}, \qquad h_j = \mathrm{dist}(j,0) \leq n/2, \tag{5.18}$$

where C is independent of j and of all sufficiently large n. Analyticity of $z(\lambda)$ near the simple off-spectrum eigenvalue and the eigenvalue estimate already proved give

$$z_n = z_\infty + O(\rho_\infty^n), \qquad \frac{|z_n|}{\rho_\infty} \leq 1 + C\rho_\infty^{n-1}.$$

It follows uniformly for $h_j \leq n/2$ that

$$\left(\frac{|z_n|}{\rho_\infty}\right)^{h_j} \leq \exp(Ch_j\rho_\infty^{n-1}) \leq \exp(Cn\rho_\infty^{n-1}/2) = O(1).$$

For large n, $|z_n| < 1$ and $n - h_j \geq h_j$, so the wrapped term in (5.18) is no larger than its first term; also $1 - |z_n|^n$ is bounded below. Thus every component is bounded by $C\rho_\infty^{h_j}$. The local coefficients converge componentwise to the nonzero infinite-volume resolvent vector. Choose a fixed site and component whose limiting coefficient is nonzero. That coefficient stays bounded away from zero, while the geometric majorant above bounds the entire unnormalized ℓ^2 norm from above. Hence this norm is bounded above and below by positive constants. Normalization therefore changes the component bound only by a fixed factor, and squaring the two components proves the squared site-amplitude bound $C\rho_\infty^{2h_j}$.

Compact case. In the compact case, the Laurent-polynomial Fourier vector displayed in the proof of Proposition 5.6 is an exact finite-ring eigenvector for every n (with the two neighbors identified when $n = 2$), proving the last assertion. ■

6. Fixed-rate spectral gap

At fixed $\omega > 0$, the first spatial mode has decay of order n^{-2} and reflection protects one of its parity components. The local pole calculation identifies the protected and affected candidate roots; the right-edge estimates below show that no other root has a larger real part. Section 7 treats the distinct bulk-critical regime in which both tumbling rates scale as n^{-1}.

6.1. Gap definition

Lemma 6.1 *(Irreducibility)***.** For every $n \geq 2$, $\omega > 0$, and $\omega_0 \geq 0$, the directed transition graph of $L_n(\omega, \omega_0)$ is strongly connected. Consequently zero is an algebraically simple eigenvalue of $L_n(\omega, \omega_0)$.

Proof. Choose a bulk site $r \neq 0$. Starting from any state, repeated hops in its current orientation reach r. At r the orientation can be retained or reversed, the latter transition having rate $\omega > 0$. Repeated hops in the selected orientation then reach any prescribed target site with the prescribed orientation. Thus every state communicates with every other state, including when $\omega_0 = 0$.

For every $t > 0$, each such finite path has positive probability of being completed before time t, so every entry of e^{tL_n} is positive. This matrix is stochastic. The Perron–Frobenius theorem therefore makes its eigenvalue one algebraically simple. A nontrivial Jordan chain of L_n at zero would exponentiate to a nontrivial Jordan chain of e^{tL_n} at one, which is impossible. Hence zero is algebraically simple. ■

Definition 6.2 (Spectral gap)**.** For $\omega > 0$ and $n \geq 2$, Lemma 6.1 makes zero a simple eigenvalue. Define

$$\gamma_n(\omega, \omega_0) = -\max\{\Re\lambda : \lambda \in \operatorname{spec} L_n(\omega, \omega_0),\ \lambda \neq 0\}.$$

Put $q = 2\pi/n$ and define the protected first-mode decay rate

$$\gamma_n^{\text{hom}}(\omega) = -\Re\lambda_1^+ = 1 - \cos q + \omega - \Re\sqrt{\omega^2 - \sin^2 q}, \tag{6.1}$$

where the principal square root is used. For each fixed $\omega > 0$, the radicand is nonnegative for all sufficiently large n, and this rate then equals the homogeneous spectral gap. No such identity is asserted for small n, where the first-mode roots can be nonreal and the tumble mode can be slower.

Proposition 6.3 *(Spectral gap and semigroup decay)***.** Let π_n be the stationary probability vector, let $\mathbf{1}$ be the all-ones column vector, and put $\Pi_n = \pi_n \mathbf{1}^\mathsf{T}$. Then $\gamma_n > 0$, and Π_n is the spectral projection of L_n at zero. Let r_n be the largest size of a Jordan block among the eigenvalues satisfying $\Re\lambda = -\gamma_n$. For every submultiplicative matrix norm there is a constant C_n such that, for $t \geq 0$,

$$e^{-\gamma_n t} \leq \|e^{tL_n} - \Pi_n\| \leq C_n(1 + t^{r_n - 1})e^{-\gamma_n t}. \tag{6.2}$$

Consequently

$$\lim_{t\to\infty} \frac{1}{t} \log\|e^{tL_n} - \Pi_n\| = -\gamma_n. \tag{6.3}$$

If $r_n > 1$ and $(v_0, \ldots, v_{r_n - 1})$ is a corresponding Jordan chain, ordered by $(L_n - \lambda I)v_0 = 0$ and $(L_n - \lambda I)v_j = v_{j-1}$, then

$$e^{tL_n} v_{r_n - 1} = e^{\lambda t} \sum_{k=0}^{r_n - 1} \frac{t^k}{k!} v_{r_n - 1 - k}. \tag{6.4}$$

Thus γ_n is the asymptotic exponential decay rate; the Jordan classification determines possible polynomial factors, while transient constants and total-variation mixing times are not specified by γ_n alone.

Proof. Lemma 6.1 shows that e^{sL_n} is a positive stochastic matrix for every $s > 0$. The strict Perron–Frobenius theorem places every eigenvalue of e^{sL_n} other than its simple eigenvalue one strictly inside the unit circle. By spectral mapping, $e^{s\lambda}$ is an eigenvalue of e^{sL_n} for every $\lambda \in \operatorname{spec} L_n$; hence every nonzero eigenvalue of L_n has negative real part and $\gamma_n > 0$.

The normalization $\mathbf{1}^{\mathsf{T}}\pi_n = 1$, together with $L_n\pi_n = 0$ and $\mathbf{1}^{\mathsf{T}}L_n = 0$, identifies the rank-one spectral projection at zero as Π_n. Put L_n into Jordan normal form on the complementary generalized eigenspaces. A block of size r at λ contributes $e^{\lambda t}\sum_{k=0}^{r-1} t^k N^k/k!$. The blocks on the spectral line $\Re\lambda = -\gamma_n$ therefore give the upper bound in (6.2); polynomial factors from blocks strictly to the left are absorbed into its constant. The spectrum of $e^{tL_n} - \Pi_n$ consists of zero and the numbers $e^{t\lambda}$ for $\lambda \neq 0$, so its spectral radius is $e^{-\gamma_n t}$ and is bounded above by every submultiplicative matrix norm. This gives the lower bound. Taking logarithms proves (6.3), and the Jordan-block exponential is exactly (6.4). ∎

6.2. Fixed-rate asymptotics

Theorem 6.4 *(Fixed-rate gap asymptotics)***.** Fix $\omega > 0$ and $\omega_0 \geq 0$.

1. If $\omega_0 \leq \omega$, then for every sufficiently large n,

$$\gamma_n(\omega,\omega_0) = \gamma_n^{\mathrm{hom}}(\omega). \tag{6.5}$$

2. If $\omega_0 > \omega$, then

$$\gamma_n(\omega,\omega_0) = \gamma_n^{\mathrm{hom}}(\omega) - \frac{4\pi^2(\omega_0-\omega)(1+\omega)}{\omega^2(1+\omega_0)}\, n^{-3} + O(|\omega_0-\omega| n^{-4}). \tag{6.6}$$

3. In both cases,

$$\begin{aligned}\gamma_n^{\mathrm{hom}}(\omega) = {} & \frac{2\pi^2(1+\omega)}{\omega}\, n^{-2} \\ & - 16\pi^4\left(\frac{1}{24} + \frac{1}{6\omega} - \frac{1}{8\omega^3}\right) n^{-4} + O(n^{-6}).\end{aligned} \tag{6.7}$$

The errors are uniform when (ω,ω_0) ranges over a compact set with ω bounded away from zero. The root selection is uniform through $\omega_0 = \omega$, where the two parity roots coincide.

The selection mechanism is visible directly in the two parity sectors. Reflection fixes the protected first-mode root at its homogeneous value. The affected companion moves away from that value at order n^{-3}: a slower defect moves it to the left and leaves the protected root in control, whereas a faster defect moves it toward zero and selects the affected root. Thus a local rate change alters the first correction but not the leading fixed-rate decay scale.

The proof occupies the next two subsections. The local Green estimate locates the affected companion of the protected first-mode root; the right-edge separation then excludes every other extended, band-edge, or localized eigenvalue.

6.3. Fixed-rate Green bounds and root separation

Lemma 6.5 *(Fixed-rate regular Green estimate)***.** Let $K \subset (0,\infty)\times[0,\infty)$ be compact and let $A > 0$. For $(\omega,\omega_0) \in K$, put

$$p_n = \lambda_1^+, \qquad R_n = -\frac{\omega - r_1}{2r_1}, \qquad r_1 = \sqrt{\omega^2 - \sin^2(2\pi/n)}.$$

For all sufficiently large n, define the regular part on $|\lambda - p_n| \le An^{-3}$ by

$$G_n^{\mathrm{reg}}(\lambda) = G_n(\lambda) - \frac{2R_n}{n(\lambda - p_n)}.$$

There is $C = C(K, A)$ such that throughout this disk

$$\left| G_n^{\mathrm{reg}}(\lambda) - \frac{1}{2(1+\omega)} \right| \le \frac{C}{n}, \qquad |(G_n^{\mathrm{reg}})'(\lambda)| \le Cn. \tag{6.8}$$

Proof. Write $q_k^* = 2\pi \min(k, n-k)/n$. Choose $\eta > 0$, uniformly for $(\omega, \omega_0) \in K$, so that $|\sin q| \le \omega/2$ for $0 \le q \le \eta$. On this interval put

$$p^{\pm}(q) = \cos q - 1 - \omega \pm \sqrt{\omega^2 - \sin^2 q}, \qquad R^+(q) = -\frac{\omega - \sqrt{\omega^2 - \sin^2 q}}{2\sqrt{\omega^2 - \sin^2 q}}.$$

Taylor's theorem with remainder, applied uniformly on the compact ω-projection of K, gives constants $c, C > 0$ such that

$$cq^2 \le -p^+(q) \le Cq^2, \qquad |R^+(q)| \le Cq^2, \tag{6.9}$$

$$|p^+(q_k^*) - p^+(q_1^*)| \ge c(k^2-1)n^{-2} \qquad (2 \le k \le \eta n/(2\pi)). \tag{6.10}$$

For the last inequality, regard $p^+(q)$ as a function of q^2; its derivative at zero is $-(1+\omega)/(2\omega)$ and, after decreasing η, its modulus is bounded below by a positive constant. The fast root $p^-(q)$ stays a fixed distance from zero on $[0, \eta]$. For $\eta \le q \le \pi$, both roots stay a fixed distance from zero: the real-part dissipation of the homogeneous Fourier block can vanish only at $q = 0$ in the density direction, and the Heine–Borel theorem applied to the remaining compact parameter set supplies a uniform positive distance.

For an interior paired wave number with distinct roots, partial fractions give

$$\frac{\lambda + 1 - \cos q_k}{D_k(\lambda)} = \frac{R^+(q_k^*)}{\lambda - p^+(q_k^*)} + \frac{1 - R^+(q_k^*)}{\lambda - p^-(q_k^*)}.$$

Let S_n be the convex hull of zero and $\{|\lambda - p_n| \le An^{-3}\}$. Every point of S_n lies to the right of the second slow pole by at least $c(k^2-1)n^{-2}$ at index $k \ge 2$, after decreasing c and increasing n. Hence equations (6.9)–(6.10) bound the derivative of the regular slow-pole sum throughout S_n by

$$Cn \sum_{k=2}^{\infty} \frac{k^2}{(k^2-1)^2} \le C'n.$$

For $q_k^* \le \eta$, every fast-pole denominator stays uniformly away from S_n. For $q_k^* \ge \eta$, including a coalesced pair with $\omega = |\sin q_k|$, use the unreduced summand in (4.3). The compactness argument above gives $|D_k(\lambda)| \ge c > 0$ on S_n, uniformly in this range. The numerator, D_k, and D_k' are uniformly bounded there, so direct differentiation of $(\lambda + 1 - \cos q_k)/D_k(\lambda)$ is $O(1)$. After the $1/n$ normalization, each corresponding paired contribution to G_n is therefore $O(n^{-1})$, without dividing by the difference of the two roots. There are at most $n/2$ such pairs. The two endpoint contributions have derivatives $O(n^{-1})$ because their poles stay a fixed distance from S_n. Thus these terms have total derivative $O(1)$. This proves the derivative bound in (6.8).

At $\lambda = 0$, every unpaired nonzero-wave Fourier summand is exactly $1/[2(1+\omega)]$. The $q = 0$ term and the removed first-pole term each change the average by $O(n^{-1})$, using (6.9). Hence $G_n^{\mathrm{reg}}(0) = 1/[2(1+\omega)] + O(n^{-1})$. Finally $|p_n| = O(n^{-2})$, so integration of the derivative bound along the segment from zero to the displayed disk proves the first bound in (6.8). ∎

Lemma 6.6 *(Protected first mode and affected pole)***.** For every n, the protected sector contains λ_1^+ and $\Re\lambda_1^+ = -\gamma_n^{\mathrm{hom}}$. Whenever $|\sin(2\pi/n)| \le \omega$, this root is real and $\lambda_1^+ = -\gamma_n^{\mathrm{hom}}$. Let $K \subset (0,\infty)\times[0,\infty)$ be compact. Uniformly for $(\omega,\omega_0) \in K$ as $n \to \infty$, the affected root associated with the same homogeneous pole is

$$\lambda_{1,-} = \lambda_1^+ + \frac{\delta(1+\omega)}{\omega^2(1+\omega_0)}\frac{q^2}{n} + O(|\delta|n^{-4}). \tag{6.11}$$

Proof. The sector assertion is Theorem 4.4; the two equalities follow from Lemma 3.2 and Definition 6.2. Near λ_1^+, the two terms $k = 1, n-1$ in the full Fourier Green function have residue

$$R_1 = \frac{\lambda_1^+ + 1 - c_1}{\lambda_1^+ - \lambda_1^-} = -\frac{\omega - r_1}{2r_1}, \qquad r_1 = \sqrt{\omega^2 - \sin^2 q}.$$

Thus

$$G_n(\lambda) = \frac{2R_1}{n(\lambda - \lambda_1^+)} + G_n^{\mathrm{reg}}(\lambda).$$

If $\delta = 0$, the affected root equals the protected root and the formula follows. Suppose $\delta \ne 0$ and put

$$M(\lambda) = 1 + 2\delta G_n^{\mathrm{reg}}(\lambda), \qquad M_0 = \frac{1+\omega_0}{1+\omega}.$$

Lemma 6.5 gives, uniformly on every fixed $O(n^{-3})$ disk about λ_1^+,

$$M(\lambda) = M_0 + O(|\delta|n^{-1}), \qquad M'(\lambda) = O(|\delta|n).$$

The pole-cleared secular equation is

$$h_n(\lambda) := (\lambda - \lambda_1^+)M(\lambda) + \frac{4\delta R_1}{n} = 0.$$

Its affine comparison has the zero

$$\lambda_* - \lambda_1^+ = -\frac{4\delta R_1}{nM_0} = O(|\delta|n^{-3}).$$

On a circle of radius $C|\delta|n^{-4}$ about λ_*, the difference between h_n and $M_0(\lambda - \lambda_*)$ is at most $C_1\delta^2 n^{-4} + O(\delta^2 n^{-5})$. Since δ ranges over a fixed bounded set, choosing C larger than $C_1|\delta|/M_0$ makes the affine term strictly dominant. Rouché's theorem gives a unique root in this circle. Therefore

$$\lambda_{1,-} - \lambda_1^+ = \frac{2\delta(\omega - r_1)}{nr_1}\frac{1}{1+\delta/(1+\omega)} + O(|\delta|n^{-4}),$$

where the constant is uniform on the compact parameter sets of the lemma. The pole-cleared equation has real coefficients, and both λ_1^+ and λ_* are real. The disk used above is invariant under complex conjugation, so uniqueness makes $\lambda_{1,-}$ real. For $\delta \neq 0$, the displayed leading coefficient of δ is positive and of order n^{-3}, whereas the remainder is $O(|\delta|n^{-4})$. Consequently the displacement has the sign of δ for all sufficiently large n. Finally $(\omega - r_1)/r_1 = q^2/(2\omega^2) + O(q^4)$, which proves (6.11). ■

Lemma 6.7 *(Fixed-rate threshold Green bounds)***.** Fix a compact $K \subset (0,\infty)\times[0,\infty)$, put $D = (1+\omega)/(2\omega)$ and $g_0 = 1/[2(1+\omega)]$, and let $p_n = \lambda_1^+$. The following estimates hold uniformly for $(\omega,\omega_0) \in K$.

1. For every $M < \infty$ there is C_M, independent of A and n, with the following property: for every fixed $A \geq 1$ and all $n \geq N(K,M,A)$, whenever $\lambda = \zeta/n^2$, $|\zeta| \leq M$, and

$$\Re\lambda \geq \Re p_n - \pi^2 D n^{-2}, \qquad |\lambda - p_n| \geq A n^{-3},$$
$$|G_n(\lambda) - g_0| \leq \frac{C_M}{A} + \frac{C_M}{n}. \tag{6.12}$$

2. For every fixed $C > 0$, if $\lambda_n \to 0$ satisfies $-Cn^{-2} \leq \Re\lambda_n \leq 0$ and $n^2|\lambda_n| \to \infty$, then

$$G_n(\lambda_n) = g_0 + o(1). \tag{6.13}$$

Proof. Use the exact identities

$$x - 1 = \frac{\lambda(\lambda+2\omega)}{2a}, \qquad \lambda + 1 - x = \frac{\lambda(\lambda+2)}{2a}. \tag{6.14}$$

For $\lambda = \zeta/n^2$, choose either normalized local branch of $w_n = n \operatorname{arccosh} x$ satisfying $w_n/n \to 0$; the two such branches differ by sign. The products used below are invariant under that sign change. Taylor's theorem on $|\zeta| \leq M$ and (6.14) give

$$w_n^2 = \frac{\zeta}{D} + O_{K,M}(n^{-2}), \tag{6.15}$$
$$\frac{\lambda+1-x}{2a\sinh(w_n/n)} = \frac{w_n}{4\omega(1+\omega)n} + O_{K,M}(|w_n|n^{-3}). \tag{6.16}$$

Here the quotients at $w_n = 0$ mean their analytic continuations. The first line in particular bounds $|w_n|$ uniformly. To verify the second line without dividing by a small w_n, divide the two identities in (6.14) and use $(\cosh t - 1)/\sinh t = \tanh(t/2)$. This gives the exact factorization

$$\frac{\lambda+1-x}{2a\sinh(w_n/n)} = \frac{\lambda+2}{2a(\lambda+2\omega)}\tanh\Big(\frac{w_n}{2n}\Big).$$

Uniformly on K and $|\zeta| \leq M$, the prefactor on the right is $1/[2\omega(1+\omega)] + O_{K,M}(n^{-2})$, while Taylor's theorem gives $\tanh(w_n/(2n)) = w_n/(2n) + O_{K,M}(|w_n|^3 n^{-3})$. The uniform bound on $|w_n|$ proves (6.16), including at $w_n = 0$.

Since $z^n = e^{-w_n}$, the exact finite Green formula becomes

$$G_n(\zeta/n^2) = g_0 + \frac{w_n}{4\omega(1+\omega)n}\coth(w_n/2) + E_n, \tag{6.17}$$

where

$$|E_n| \,\le\, C_{K,M} n^{-2} + C_{K,M} n^{-3} |w_n \coth(w_n/2)|.$$

Let $q_\ell \,=\, 2\pi\ell/n$ for $1 \,\le\, \ell \,\le\, n/2$. Uniform Taylor expansion on the compact ω-projection of K gives

$$p^+(q_\ell) \,=\, -Dq_\ell^2 + O_K(n^{-4}) \qquad (\ell \,=\, 1, 2).$$

In particular,

$$\begin{aligned} p^+(q_2) - p_n \,&=\, -12\pi^2 D n^{-2} + O_K(n^{-4}), \\ \Re p_n - \pi^2 D n^{-2} \,&=\, -5\pi^2 D n^{-2} + O_K(n^{-4}). \end{aligned}$$

Hence $p^+(q_2)$ lies strictly to the left of the stated half-plane for all sufficiently large n, uniformly on K. As in the proof of Lemma 6.5, $p^+(q)$ is a strictly decreasing function of q^2 on a fixed interval $0 \,\le\, q \,\le\, \eta$, while the fast branch there and both branches for $\eta \,\le\, q \,\le\, \pi$ stay a fixed distance to the left of zero. These facts exclude every remaining homogeneous pole from the half-plane. The zero density eigenvalue is removable in G_n, since its $q \,=\, 0$ summand reduces to $1/[n(\lambda + 2\omega)]$. The only nonzero pole in the stated scaled half-plane has $w_n \,=\, 2\pi\mathrm{i}$ up to sign and is exactly $\lambda \,=\, p_n$. The derivative of the analytic map $\zeta \mapsto w_n$ at this pole is bounded above and below away from zero. The complex inverse-function theorem, uniformly on the compact parameter set, therefore gives $|w_n - 2\pi\mathrm{i}| \,\ge\, cA/n$ outside $|\lambda - p_n| \,<\, An^{-3}$. On a fundamental strip,

$$|\coth(w/2)| \,\le\, C\left(1 + \operatorname{dist}(w, 2\pi\mathrm{i}\mathbb{Z})^{-1}\right); \tag{6.18}$$

indeed, subtract the nearest $2\pi\mathrm{i}\ell$, use $\sinh(w/2) \,=\, w/2 + O(w^3)$ for $|w| \,\le\, 1$, and use continuity on the remaining closed strip together with $\coth(w/2) \to \pm 1$ as $\Re w \to \pm\infty$. If $|w_n| \,\le\, \eta$ for a sufficiently small fixed η, then $w_n \coth(w_n/2)$ is bounded by its analytic continuation at zero. Thus the correction term in (6.17) is $O_{K,M}(n^{-1})$ and $E_n \,=\, O_{K,M}(n^{-2})$ there. On the complementary region $|w_n| \,>\, \eta$, the uniform upper bound on $|w_n|$, the exclusion of the first pole, and (6.18) give

$$|\coth(w_n/2)| \,\le\, C_{K,M}\left(1 + \frac{n}{A}\right).$$

The correction term is consequently bounded by $C_{K,M}/n + C_{K,M}/A$, and the new remainder bound gives $|E_n| \,\le\, C_{K,M} n^{-2} + C_{K,M}/(An^2)$. This proves (6.12), including the removable point $w_n \,=\, 0$.

For item 2, first observe that λ_n is not a homogeneous pole for all sufficiently large n. Choose $\eta \,>\, 0$ as in the proof of Lemma 6.5. The fast branch on $[0, \eta]$ and both branches on $[\eta, \pi]$ stay uniformly away from zero. If a remaining pole were equal to λ_n, it would have the form $p^+(q_k^*)$; then (6.9) and $\Re\lambda_n \,\ge\, -Cn^{-2}$ would give $q_k^* \,=\, O(n^{-1})$ and hence $|\lambda_n| \,=\, O(n^{-2})$, contrary to $n^2|\lambda_n| \to \infty$. The real-part assumptions now imply $|\Im\lambda_n|/|\lambda_n| \to 1$. By the first identity in (6.14), $x - 1$ lies, for large n, in one of two closed sectors about the positive or negative imaginary axis. The expansion

$$\operatorname{arccosh}(1 + u) \,=\, \sqrt{2u}\,(1 + O(u))$$

on those sectors gives $\Re \operatorname{arccosh} x \ge c\sqrt{|\lambda_n|}$. Hence

$$|z(\lambda_n)|^n \le \exp(-cn\sqrt{|\lambda_n|}) \longrightarrow 0.$$

Moreover, the two identities in (6.14) give $|(\lambda_n + 1 - x)/s| \le C\sqrt{|\lambda_n|}$. Substitution in (5.15) now proves (6.13). ■

Lemma 6.8 *(Right-edge root separation)***.** Fix $\omega > 0$ and $\omega_0 \ge 0$, and put $D = (1+\omega)/(2\omega)$. There is N such that, for $n \ge N$, every nonzero eigenvalue other than the protected and affected roots associated with $q = \pm 2\pi/n$ satisfies

$$\Re\lambda \le \Re\lambda_1^+ - \pi^2 D n^{-2}. \tag{6.19}$$

The constants can be chosen uniformly when (ω, ω_0) ranges over a compact set with ω bounded away from zero.

Proof. Right-edge window. For the uniform assertion, fix a compact $K \subset (0,\infty) \times [0,\infty)$ containing the given parameter pair; the pointwise assertion follows from the same argument. All parameter-dependent objects below are evaluated at the current pair in K. Put

$$\Omega_n = \left\{\lambda : \Re\lambda \ge \Re\lambda_1^+ - \pi^2 D n^{-2}\right\}.$$

Protected-sector exclusion. The protected roots are explicit. Their slow branch satisfies, uniformly for small real q,

$$\lambda^+(q) = -Dq^2 + O(q^4). \tag{6.20}$$

Choose $q_{\mathrm{loc}} > 0$ so that the remainder is at most $Dq^2/4$ for $|q| \le q_{\mathrm{loc}}$. The discrete Fourier modes with $2\pi/n < |q| \le q_{\mathrm{loc}}$ are then to the left of Ω_n, while continuity puts both branches with $q_{\mathrm{loc}} \le |q| \le \pi$ a fixed distance into the left half-plane. The fast small-q branch is also separated from zero by a fixed amount. Thus, apart from zero, only the protected root associated with $q = \pm 2\pi/n$ lies in Ω_n once n is large. We prove the same exclusion for the affected sector.

Affected secular equation. If $\delta = 0$, that sector is homogeneous and the conclusion follows from (6.20). Assume $\delta \ne 0$ and set

$$f_n(\lambda) = 1 + 2\delta G_n(\lambda), \qquad g_0 = \frac{1}{2(1+\omega)}, \qquad M_0 = 1 + 2\delta g_0 = \frac{1+\omega_0}{1+\omega} > 0. \tag{6.21}$$

When parameters vary below, Ω_n, f_n, g_0, and M_0 are evaluated at the current parameter pair. In a neighborhood of zero, every affected root is a zero of f_n. Indeed, the residue at every slow homogeneous pole is nonzero, so multiplication by $H_{n,-}^{(0)}$ cancels the pole of f_n but does not leave a zero there.

Exclusion away from the origin. We first exclude, uniformly on K, sequences of affected zeros that do not tend to zero. The protected roots and the homogeneous case $\delta = 0$ were handled above. If the claimed exclusion failed for the remaining affected roots, there would be $n_j \to \infty$, parameters $(\omega^{(j)}, \omega_0^{(j)}) \in K$ with $\delta^{(j)} = \omega_0^{(j)} - \omega^{(j)} \ne 0$, and affected zeros

$\lambda_j \in \Omega_{n_j}$, with the window evaluated at that pair, such that $|\lambda_j| \geq \epsilon > 0$. Gershgorin's theorem places these eigenvalues in a fixed disk and in the closed left half-plane. Pass to a subsequence such that

$$(\omega^{(j)}, \omega_0^{(j)}) \longrightarrow (\omega_*, \omega_{0,*}), \qquad \lambda_j \longrightarrow \lambda_* \neq 0.$$

Since the boundary of Ω_{n_j} tends to the imaginary axis, $\Re\lambda_* = 0$. For a bulk Bloch vector at wave number q, the real part of its quadratic form is the negative of $(1-\cos q)$ times its squared norm, together with the nonnegative tumbling form. Equality therefore forces $q = 0$ and equal orientation components, and the corresponding eigenvalue is zero. Thus Σ_{ω_*} meets the imaginary axis only at zero.

Choose a closed disk U about λ_* disjoint from zero, Σ_{ω_*}, and the point $a = 0$. By continuity of the bulk symbol and compactness, after shrinking U and increasing j, it is disjoint from $\Sigma_{\omega^{(j)}}$ and $a = 0$ as well. The local branches $z(\lambda; \omega^{(j)})$ can then be chosen jointly on U and satisfy $\sup_U |z(\lambda; \omega^{(j)})| \leq \rho < 1$. Equation (5.16) gives the uniform estimate

$$\sup_{\lambda \in U} |G_{n_j}(\lambda; \omega^{(j)}) - g_\infty(\lambda; \omega^{(j)})| \leq C_U \frac{\rho^{n_j}}{1-\rho^{n_j}} \longrightarrow 0.$$

For large j, the homogeneous odd factor is nonzero on U, so the affected polynomial equation there is the secular equation. It therefore passes to

$$1 + 2(\omega_{0,*} - \omega_*) g_\infty(\lambda_*; \omega_*) = 0.$$

If $\omega_{0,*} = \omega_*$ this equation is already impossible. Otherwise it produces an ℓ^2 eigenvector u of the limiting infinite defective generator with eigenvalue λ_*. This is impossible because

$$-\Re\langle u, L_\infty u\rangle = \frac{1}{2}\sum_{j,\sigma} |u_{j+1}^\sigma - u_j^\sigma|^2 + \sum_j \omega_j |u_j^+ - u_j^-|^2. \tag{6.22}$$

Here (6.22) follows by expanding the inner product, shifting the two hopping sums, and pairing the two tumbling terms at each site. These operations are valid first for finitely supported vectors and then for all $u \in \ell^2$ by density and boundedness of L_∞. Equality in (6.22) forces an ℓ^2 vector to vanish. Consequently every sequence of affected zeros in Ω_n tends to zero, uniformly for parameters in K.

Scaled exhaustion near the origin. It remains to exhaust these affected zeros. On the $\zeta = n^2\lambda$ scale, the limiting first pole is $-4\pi^2 D(\omega)$, where $D(\omega) = (1+\omega)/(2\omega)$. Its range over the compact ω-projection of K is compact, and for each ω it stays uniformly separated from the corresponding limiting second pole. Choose a uniform neighborhood radius smaller than half this separation, and then choose M_* large enough to contain all the resulting first-pole neighborhoods. Let C_{M_*} be the uniform constant from item 1 of Lemma 6.7, and put

$$\Delta_K = \max_K |\omega_0 - \omega|, \qquad m_K = \min_K \frac{1+\omega_0}{1+\omega} > 0.$$

Choose A so large that $2\Delta_K C_{M_*}/A < m_K/4$. Only after these choices do we increase n. Suppose, to the contrary, that there were a sequence $n \to \infty$, parameters $(\omega_n, \omega_{0,n}) \in K$

with $\delta_n = \omega_{0,n} - \omega_n \neq 0$, and affected zeros $\lambda_n \in \Omega_n$ outside $|\lambda - \lambda_1^+| < An^{-3}$. Gershgorin's theorem gives $\Re\lambda_n \leq 0$, and the preceding paragraph gives $\lambda_n \to 0$. The definition of Ω_n and (6.20) give $-Cn^{-2} \leq \Re\lambda_n \leq 0$. Pass to a subsequence according as $\zeta_n = n^2\lambda_n$ is bounded or unbounded. In the bounded case, pass once more so that the parameters converge and $\zeta_n \to \zeta_*$. If ζ_* is not the corresponding limiting first pole, (6.17)–(6.18) give $G_n(\lambda_n) \to g_0$. If it is the limiting first pole, then ζ_n lies in its chosen neighborhood for all sufficiently large n, and the first assertion of that lemma and the choice of A give $|G_n(\lambda_n) - g_0| \leq C_{M_*}/A + o(1)$. In both alternatives, $|f_n(\lambda_n)| > m_K/2$ for large n, a contradiction. In the unbounded case, the second assertion of the lemma gives $G_n(\lambda_n) = g_0 + o(1)$ and the same contradiction. Hence, for all sufficiently large n, every affected zero in Ω_n lies in the first-pole disk, with a threshold uniform on K.

First-pole count. Finally, on the circle $|\lambda - \lambda_1^+| = An^{-3}$ write

$$G_n(\lambda) = \frac{2R_1}{n(\lambda - \lambda_1^+)} + G_n^{\mathrm{reg}}(\lambda).$$

Lemma 6.5 and (6.21) give $1 + 2\delta G_n^{\mathrm{reg}} = M_0 + O(n^{-1})$. After multiplication by $\lambda - \lambda_1^+$, the secular equation becomes

$$(\lambda - \lambda_1^+) f_n(\lambda) = M_0(\lambda - \lambda_1^+) + \frac{4\delta R_1}{n} + O(n^{-1}|\lambda - \lambda_1^+|). \tag{6.23}$$

Here $R_1 = O(n^{-2})$. Enlarge A, if necessary, so that the zero of the first two terms lies strictly inside the circle. On the circle the last term is $O(n^{-4})$, whereas the first two terms have modulus bounded below by $c_A n^{-3}$. Rouché's theorem therefore counts exactly one zero in the disk. It is the affected root of Lemma 6.6. Thus no other affected root occurs in Ω_n, which proves the assertion.

All estimates above are uniform on the stated compact parameter sets: D, δ, and the generator norms are bounded there, while M_0 has a positive minimum. ∎

6.4. Gap selection and fixed-rate consequences

Proof of Theorem 6.4. Taylor expansion of (6.1) gives (6.7). Lemma 6.6 shows that, uniformly across $\delta = 0$, the affected first root moves toward zero precisely when $\delta > 0$. Lemma 6.8 shows that no other extended, band-edge, or localized root can compete with this first pair. Therefore the protected root has maximal real part when $\delta \leq 0$, proving (6.5), while the affected root has maximal real part when $\delta > 0$. Substitution of $q^2 = 4\pi^2 n^{-2}$ in (6.11) then gives (6.6). ∎

Corollary 6.9 *(Fixed ratio, vanishing ratio, and a slow-site scaling)***.** Let the bulk rate $\omega > 0$ be fixed.

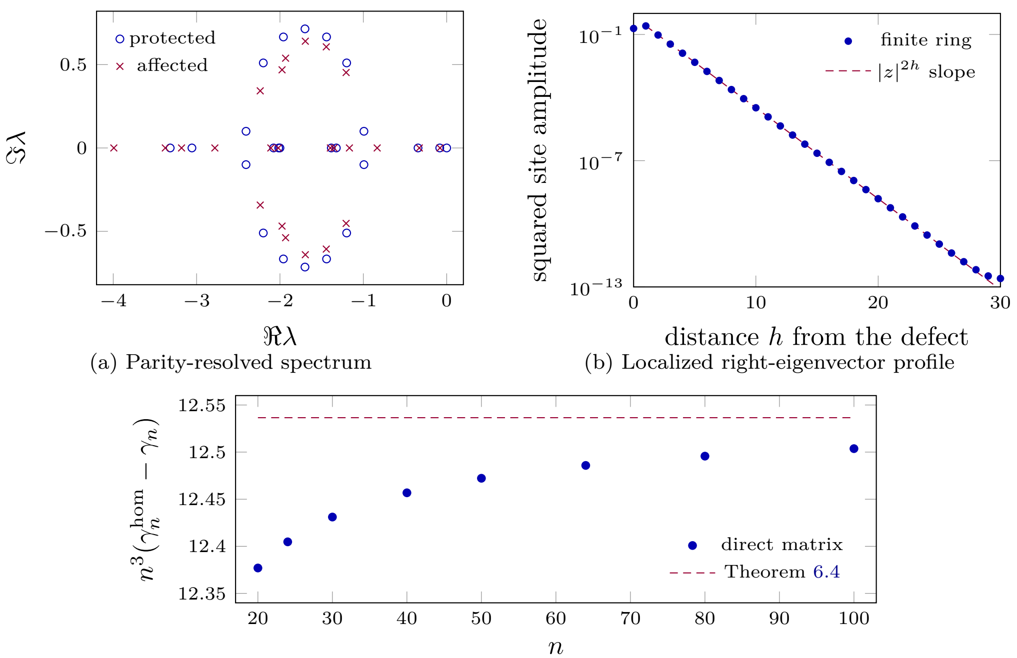
1. If $\omega_0/\omega \to \rho \in (0, \infty)$, equations (6.5)–(6.7) apply. For $\rho < 1$ the protected branch is selected; for $\rho > 1$ the affected branch and its n^{-3} term are selected. At the boundary $\rho = 1$, the protected root is selected whenever $\omega_0 \leq \omega$ and the affected root whenever $\omega_0 > \omega$, with the expansion uniform across equality.

2. If $\omega_0/\omega \to 0$, then (6.5) holds for every sufficiently large n.

3. In particular, if $n\omega_0 \to \beta$ with $\beta \geq 0$, then (6.5) still holds. A single slow tumbling site therefore has no separate $\omega_0 \asymp n^{-1}$ gap crossover when the bulk rate is fixed.

Proof. In items 2 and 3 one has $\omega_0 < \omega$ for every sufficiently large n, so Theorem 6.4 applies. In item 1 the convergence of ω_0/ω places the n-dependent pairs (ω, ω_0) in one compact parameter set. The compact-uniform statement in Theorem 6.4 therefore permits this n-dependence, and its sign test gives the asserted branch selection, including crossings through equality. ■

Figure 2 illustrates three parts of the exact analysis with direct finite matrices: the protected/affected parity split, the infinite-support localized tail, and the fixed-rate gap correction.



(a) Parity-resolved spectrum

(b) Localized right-eigenvector profile

(c) Faster-defect gap correction

Figure 2: Representative finite-ring diagnostics. (a) The spectrum at $n = 24$, $\omega = 0.7$, and $\omega_0 = 1.4$, computed separately in the two reflection sectors. (b) The normalized squared right-eigenvector amplitude at $n = 60$, $\omega = 0.5$, and $\omega_0 = 0.8$ for the eigenvalue converging to $\lambda = -1.093854643\ldots$; the reference slope uses $z = 0.607977704\ldots$ from Corollary 5.5. This panel concerns a right eigenvector, not stationary probability mass. (c) For $\omega = 1.3$ and $\omega_0 = 2$, direct gaps approach the n^{-3} coefficient in Theorem 6.4. The plotted data are generated by `scripts/generate_figure_data.py`.

7. Bulk-critical spectral gap

When $\omega = \alpha/n$ and $\omega_0 = \beta/n$, the gap is of order n^{-1} and the limiting first-mode exceptional scale $\alpha = 2\pi$ separates two parity-selection patterns. Local pole calculations locate the candidate roots, while the right-edge estimates below exclude every other root from controlling the gap. We call this regime bulk-critical because the bulk tumbling time ω^{-1} and the ballistic traversal time of the ring are both of order n.

7.1. Asymptotic regimes and parity selection

For a real number x, write $x_+ = \max\{x, 0\}$. The scaled first wave number is denoted by $q_\star = 2\pi$. For finite n, the first homogeneous block is exceptional at

$$\alpha_{\mathrm{EP},n} = n\sin(q_\star/n) = q_\star - \frac{q_\star^3}{6n^2} + O(n^{-4}). \tag{7.1}$$

Thus $\alpha = q_\star$ is the limiting first-mode exceptional scale, not the exact finite-n coalescence. At the exact scaling $\alpha = q_\star$, the residual lattice splitting is of order n^{-2} and is retained in Theorem 7.1.

Theorem 7.1 *(Bulk-critical double scaling)***.** Set

$$\omega = \frac{\alpha}{n}, \qquad \omega_0 = \frac{\beta}{n}, \qquad q_\star = 2\pi,$$

where $\alpha > 0$ and $\beta \geq 0$ are fixed. Write $\gamma_n = \gamma_n(\alpha/n, \beta/n)$ for the gap in Definition 6.2. Away from $\alpha = q_\star$,

$$\alpha < q_\star : \quad \gamma_n = \frac{\alpha}{n} + \left[\frac{q_\star^2}{2} - 2(\alpha-\beta)_+\right] n^{-2} + O(n^{-3}), \tag{7.2}$$

$$\begin{aligned}\alpha > q_\star : \quad \gamma_n = {} & \frac{\alpha - \sqrt{\alpha^2 - q_\star^2}}{n} \\ & + \left[\frac{q_\star^2}{2} - 2(\beta-\alpha)_+ \frac{\alpha - \sqrt{\alpha^2 - q_\star^2}}{\sqrt{\alpha^2 - q_\star^2}}\right] n^{-2} + O(n^{-3}).\end{aligned} \tag{7.3}$$

At $\alpha = q_\star$,

$$\beta \leq q_\star : \quad \gamma_n = \frac{q_\star}{n} + q_\star^2\left(\frac{1}{2} - \frac{1}{\sqrt{3}}\right) n^{-2} + O(n^{-3}), \tag{7.4}$$

$$\beta > q_\star : \quad \gamma_n = \frac{q_\star}{n} - 2\sqrt{q_\star(\beta - q_\star)}\, n^{-3/2} + O(n^{-2}). \tag{7.5}$$

The constants in the $O(\cdot)$ terms may depend on the fixed pair (α, β). The side expansions (7.2)–(7.3) are for fixed $\alpha \neq q_\star$ and are not uniform as $\alpha \to q_\star$; the exact scaling $\alpha = q_\star$ is the separate regime (7.4)–(7.5). No uniformity as $\alpha \downarrow 0$ is asserted for the remainder constants or the lower bound on n. The affected parity mode controls the gap for $\beta < \alpha$ below $q_\star$ and for $\beta > \alpha$ above $q_\star$; at $\alpha = q_\star$, a faster defect produces the $n^{-3/2}$ square-root correction.

Below $q_\star$, the protected first pair is nonreal and the defect selects between equal leading real parts through an n^{-2} correction. Above $q_\star$, the first pair is real and the direction of selection reverses. At $\alpha = q_\star$ the two homogeneous roots coalesce on the leading scale, which accounts for the distinct $n^{-3/2}$ response to a faster defect.

Table 2 and Figure 3 display this selection rule. An equality case in the table means that the protected and affected parity roots coincide.

Regime	Gap-controlling parity	Gap and first correction
$\alpha < q_\star$	affected if $\beta < \alpha$; protected if $\beta > \alpha$; tie if $\beta = \alpha$	αn^{-1} with an n^{-2} correction; (7.2)
$\alpha = q_\star$, $\beta \le q_\star$	protected if $\beta < q_\star$; tie if $\beta = q_\star$	$q_\star n^{-1}$ with an n^{-2} correction; (7.4)
$\alpha = q_\star$, $\beta > q_\star$	affected	$q_\star n^{-1}$ with an $n^{-3/2}$ correction; (7.5)
$\alpha > q_\star$	protected if $\beta < \alpha$; affected if $\beta > \alpha$; tie if $\beta = \alpha$	$(\alpha - \sqrt{\alpha^2 - q_\star^2})n^{-1}$ with an n^{-2} correction; (7.3)

Table 2: Selection and correction scales in the bulk-critical limit, with $q_\star = 2\pi$.

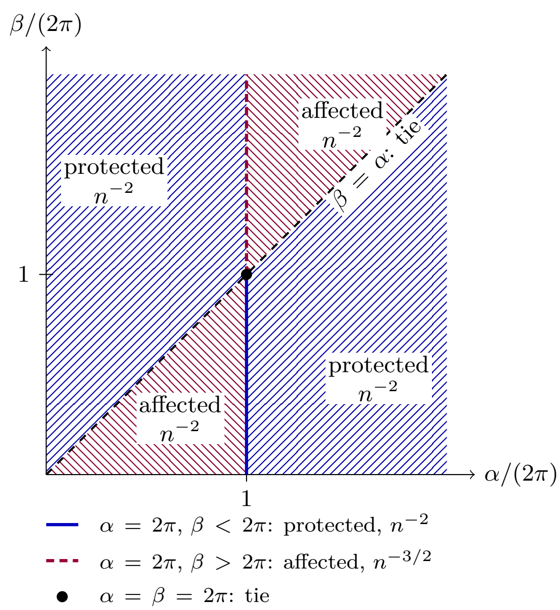


Figure 3: Gap-controlling parity and first correction scale in the bulk-critical limit. Region labels apply away from $\alpha = 2\pi$; the colored segments give the separate boundary regime. Opposite hatching distinguishes the two parities in monochrome, and the solid/dashed boundary styles distinguish the two critical segments. The displayed window is schematic: all four regions continue throughout the positive (α, β) quadrant.

The proof proceeds in four stages. Lemma 7.3 derives the local Green limits and uniform pole remainders. Lemmas 7.4 and 7.5 convert geometric pole separation into off-pole Green bounds. Lemmas 7.6– 7.9 count the pole-neighborhood roots, determine the zero-wave root, and exclude every remaining root from the spectral right edge. The final subsection expands and compares the surviving first-cluster candidates.

7.2. Critical Green function and pole neighborhoods

Definition 7.2 (Critical homogeneous poles and residues)**.** Let $\alpha > 0$, set $\omega = \alpha/n$ and $q_\star = 2\pi$, and, for an integer ℓ, put

$$\begin{aligned} p^{\pm}_{\ell,n} &= \cos(\ell q_\star/n) - 1 - \frac{\alpha}{n} \pm \sqrt{\frac{\alpha^2}{n^2} - \sin^2(\ell q_\star/n)}, \\ R^{\pm}_{\ell,n} &= \frac{p^{\pm}_{\ell,n} + 1 - \cos(\ell q_\star/n)}{p^{\pm}_{\ell,n} - p^{\mp}_{\ell,n}}. \end{aligned} \tag{7.6}$$

The square root is chosen nonnegative when its argument is nonnegative and with nonnegative imaginary part when its argument is negative. When the two poles are distinct and $1 \le \ell < n/2$, the paired waves $q = \pm \ell q_\star/n$ contribute $2R^{\pm}_{\ell,n}/\{n(\lambda - p^{\pm}_{\ell,n})\}$ for the indicated sign. The unpaired endpoint contributions are instead

$$\frac{1}{n(\lambda + 2\alpha/n)}, \qquad \frac{\varepsilon_n}{n(\lambda + 2 + 2\alpha/n)},$$

at $q = 0$ and $q = \pi$, respectively.

Lemma 7.3 *(Critical Green limit and pole remainders)***.** Set $\omega = \alpha/n$, where $\alpha > 0$ is fixed, and put $q_\star = 2\pi$. For $\nu \in \mathbb{C}$, define the meromorphic function

$$\mathcal{G}_\alpha(\nu) = \frac{1}{2}\left[1 + \frac{\nu}{\kappa}\coth\Big(\frac{\kappa}{2}\Big)\right], \qquad \kappa^2 = \nu(\nu + 2\alpha), \tag{7.7}$$

where the right-hand side is independent of the choice of square root and is interpreted by meromorphic continuation at $\kappa = 0$. The decompositions in items 2–4 are identities of meromorphic functions. At each removed pole, the difference defining the regular part is evaluated by its unique holomorphic extension. Then:

1. $G_n(\nu/n) \to \mathcal{G}_\alpha(\nu)$ locally uniformly on every compact set containing no pole of (7.7).
2. Fix $\ell \ge 1$ with $\alpha \ne \ell q_\star$, and let $p^{\pm}_{\ell,n}$ and $R^{\pm}_{\ell,n}$ be given by (7.6). There are $\eta, C > 0$ such that
$$\left| G_n(\lambda) - \frac{2R^{\pm}_{\ell,n}}{n(\lambda - p^{\pm}_{\ell,n})} \right| \le C \tag{7.8}$$
whenever $|n\lambda - np^{\pm}_{\ell,n}| \le \eta$, for all sufficiently large n.
3. If $\alpha = q_\star$, put
$$\lambda_c = \cos(q_\star/n) - 1 - q_\star/n, \qquad r_n = \sqrt{(q_\star/n)^2 - \sin^2(q_\star/n)}.$$
There are $\eta, C > 0$ and analytic functions H_n on $|n(\lambda - \lambda_c)| \le \eta$ such that
$$G_n(\lambda) = \frac{2(\lambda - \lambda_c - q_\star/n)}{n\{(\lambda - \lambda_c)^2 - r_n^2\}} + H_n(\lambda), \qquad |H_n(\lambda)| \le C. \tag{7.9}$$

4. There are $\eta, C > 0$ and analytic functions K_n on $|n\lambda + 2\alpha| \leq \eta$ such that

$$G_n(\lambda) = \frac{1}{n(\lambda + 2\alpha/n)} + K_n(\lambda), \qquad |K_n(\lambda)| \leq C. \tag{7.10}$$

Proof. Let $w_n(\nu) = n \operatorname{arccosh} x(\nu/n)$ on a local branch satisfying $w_n/n \to 0$; such normalized branches differ only by sign. The exact identities (6.14) give

$$\cosh(w_n/n) = 1 + \frac{\nu(\nu + 2\alpha)}{2n^2\{1 + (\nu + \alpha)/n\}}, \tag{7.11}$$

$$\frac{\lambda + 1 - x}{s} = \frac{\nu(2 + \nu/n)}{2n\{1 + (\nu + \alpha)/n\}\sinh(w_n/n)}, \qquad \lambda = \nu/n. \tag{7.12}$$

On a compact ν-set, Taylor's theorem applied to (7.11) yields $w_n^2 = \nu(\nu + 2\alpha) + O(n^{-1})$ uniformly. Substitution of (7.12) into the exact formula (5.15), with $(1 + z^n)/(1 - z^n) = \coth(w_n/2)$, gives the pointwise limit in item 1. The resulting expression is even and meromorphic in w_n, hence meromorphic in w_n^2, with its value at $w_n = 0$ taken by continuation. On a compact set avoiding the limiting poles, the denominators in this even expression stay uniformly separated from zero. Uniform convergence of w_n^2, together with the explicit coefficients in (7.12), therefore gives uniform convergence there. A general branch change has the form $w_n \mapsto \pm w_n + 2\pi i n m$; the sign changes in $\sinh(w_n/n)$ and $\coth(w_n/2)$ cancel, and the additive term is a period of both factors. Thus the conclusion agrees on overlapping branch neighborhoods and is independent of the normalized branch.

For item 2, the limiting poles are

$$\nu_\ell^\pm = -\alpha \pm \sqrt{\alpha^2 - \ell^2 q_\star^2}.$$

The hypothesis $\alpha \neq \ell q_\star$ makes them simple. Choose a closed disk U about one of them that contains no other limiting pole. For large n, $G_n(\nu/n)$ has exactly the corresponding pole $np_{\ell,n}^\pm$ in U, with principal part $2R_{\ell,n}^\pm/(\nu - np_{\ell,n}^\pm)$. Item 1 bounds both the Green function and this principal part on ∂U. Their difference is removable at the corresponding pole and extends analytically throughout U, so the maximum-modulus theorem bounds it on a smaller concentric disk. Replacing ν by $n\lambda$ gives (7.8).

When $\alpha = q_\star$, the two first poles lie at $\lambda_c \pm r_n$. Adding their two summands in (4.3) gives exactly the displayed rational term in (7.9). Choose a fixed disk about $-q_\star$ in the ν-plane that contains no other limiting pole. On its boundary, item 1 bounds $G_n(\nu/n)$ and direct inspection bounds the grouped term. Their difference has removable singularities at both first poles and extends analytically inside. The maximum-modulus theorem gives the asserted uniform bound on a smaller disk.

Finally, the first summand of (4.3) is exactly $1/\{n(\lambda + 2\alpha/n)\}$. The point -2α is distinct from all the limiting poles with $\ell \geq 1$. Choose a fixed disk about it in the ν-plane containing no such pole. For large n, the corresponding scaled disk contains no finite-ring pole except the $q = 0$ pole. On the boundary, item 1 bounds $G_n(\nu/n)$ and the displayed summand; their difference has a removable singularity at the endpoint pole and extends analytically to the disk. The maximum-modulus theorem on a smaller disk proves (7.10). ∎

7.3. Off-pole control

Lemma 7.4 *(Critical pole-distance conversion).* Fix $\alpha > 0$ and $R > 0$, put $\omega = \alpha/n$ and $q_\star = 2\pi$, and let w satisfy $\cosh(w/n) = x(\lambda)$ in either right-edge window below. Define the branch-invariant distances

$$d_{0,n}(w) = \operatorname{dist}(w, 2\pi\mathrm{i}n\mathbb{Z}), \qquad d_{*,n}(w) = \operatorname{dist}\big(w, 2\pi\mathrm{i}(\mathbb{Z} \setminus n\mathbb{Z})\big).$$

Thus $d_{0,n}$ measures distance to the zero-wave lattice points. In the right-edge windows below, the nonremovable tumble pole $\lambda = -2\omega$ lies outside the window, so only the removable $\lambda = 0$ branch is relevant there. The distance $d_{*,n}$ measures distance to every nonzero-wave pole. The following estimates hold for $|\lambda| \le R$.

1. Suppose $\alpha \le q_\star$. For an integer $L \ge 1$, set

   $$B = \frac{q_\star^2}{4}\{L^2 + (L+1)^2\}, \qquad \Omega_{B,n} = \{\Re\lambda \ge -\alpha/n - B/n^2\}.$$

   Delete disks of radius An^{-2} about the simple homogeneous poles in $\Omega_{B,n}$. When $\alpha = q_\star$, replace the two first-pole disks by the disk of radius $An^{-3/2}$ about their midpoint. There are $C = C(\alpha, R)$ and $C_B = C_B(\alpha, B, R)$ such that, for every fixed $A \ge 1$ and all sufficiently large n,

   $$d_{*,n}(w)^{-1} \le C_B\left(1 + \frac{n}{A}\right) + C\frac{n}{\sqrt{B}} \tag{7.13}$$

   on the remaining part of $\Omega_{B,n}$.

2. Suppose $\alpha > q_\star$. Put

   $$d_1 = \alpha - \sqrt{\alpha^2 - q_\star^2}, \qquad d_2 = \alpha - \Re\sqrt{\alpha^2 - 4q_\star^2},$$

   fix $d_{\mathrm{cut}} \in (d_1, d_2)$, and set $\Omega_{\mathrm{cut},n} = \{\Re\lambda \ge -d_{\mathrm{cut}}/n\}$. After deleting the disk of radius An^{-2} about the slow first pole, there is $C_{\mathrm{cut}} = C_{\mathrm{cut}}(\alpha, d_{\mathrm{cut}}, R)$ such that

   $$d_{*,n}(w)^{-1} \le C_{\mathrm{cut}}\left(1 + \frac{n}{A}\right) \tag{7.14}$$

   throughout the remaining part of $\Omega_{\mathrm{cut},n}$, for every fixed $A \ge 1$ and all sufficiently large n.

The thresholds for n may depend on the displayed fixed parameters and on A, but the constants in (7.13)– (7.14) do not depend on A or n.

Proof. Branch reduction. A branch change has the form $w \mapsto \pm w + 2\pi\mathrm{i}nm$. Both pole sets in the definition are invariant under this transformation, so both distances are branch independent. It is enough to consider $d_{*,n}(w) < 1$. Choose the nearest nonzero-wave representative and use the branch symmetries to write it as $2\pi\mathrm{i}\ell$, where $1 \le \ell \le \lfloor n/2 \rfloor$, and put

$$q = \frac{\ell q_\star}{n}, \qquad \theta = w - 2\pi\mathrm{i}\ell.$$

The exact factorization

$$2a\{\cosh(w/n) - \cos(\ell q_\star/n)\} \;=\; D_\ell(\lambda) \;=\; (\lambda - p^+_{\ell,n})(\lambda - p^-_{\ell,n}) \tag{7.15}$$

shows that the two zeros over this representative are precisely the two homogeneous poles of wave number q. Taylor's theorem, $|a| \le C_R$, and $|\theta| < 1$ give

$$|D_\ell(\lambda)| \;\le\; C_R\left(\frac{|\sin q|}{n}|\theta| + \frac{|\theta|^2}{n^2}\right). \tag{7.16}$$

Subcritical and critical windows. Suppose first that $\alpha \le q_\star$. The poles with indices $1 \le \ell \le L$ lie in $\Omega_{B,n}$ for all sufficiently large n. Except for $(\alpha, \ell) = (q_\star, 1)$, they are simple. On the scaled variable $\nu = n\lambda$, the limiting relation is $w^2 = \nu(\nu + 2\alpha)$, and at a simple limiting pole

$$\frac{dw}{d\nu} \;=\; \frac{\nu + \alpha}{w} \;\neq\; 0.$$

Fixed-index simple poles. For each of these fixed indices, (7.16) first gives $|D_\ell(\lambda)| \le C_B n^{-2}$. The factorization in (7.15) therefore gives

$$\min_{\sigma\in\{+,-\}} |\lambda - p^\sigma_{\ell,n}|^2 \;\le\; |(\lambda - p^+_{\ell,n})(\lambda - p^-_{\ell,n})| \;\le\; C_B n^{-2}.$$

Both poles are $O(n^{-1})$, so $|\lambda| \le C'_B n^{-1}$ and hence $\nu = n\lambda$ lies in a fixed compact set. To make the finite-n uniformity explicit, write the local branch as $w = w_n(\nu)$. Differentiating the exact relation $\cosh(w_n/n) = x(\nu/n)$ gives

$$\frac{\partial w_n}{\partial \nu} \;=\; \frac{x'(\nu/n)}{\sinh(w_n/n)}.$$

The identity

$$x(\lambda) - 1 \;=\; \frac{\lambda(\lambda + 2\omega)}{2(1 + \omega + \lambda)}$$

then gives, uniformly on each of these fixed scaled neighborhoods,

$$\frac{\partial w_n}{\partial \nu} \;=\; \frac{\nu + \alpha}{w_n} + O_B(n^{-1}).$$

For each simple limiting pole $\nu^\pm_\ell = -\alpha \pm \sqrt{\alpha^2 - \ell^2 q_\star^2}$, choose a closed disk $U^\pm_\ell$ of radius $2r_B$ in the ν-plane. Choose the finitely many disks pairwise disjoint and separated from $\nu = 0$, $\nu = -2\alpha$, and the coalesced first-pole point when $\alpha = q_\star$. Apply the analytic implicit-function theorem to

$$F_n(\nu, w) \;=\; n^2\big(\cosh(w/n) - x(\nu/n)\big).$$

On each product of $U^\pm_\ell$ with a sufficiently small fixed disk about $2\pi i\ell$, F_n and its derivatives converge uniformly to $F_\infty(\nu, w) = \frac{1}{2}\{w^2 - \nu(\nu + 2\alpha)\}$. Since $\partial_w F_\infty = 2\pi i\ell \neq 0$ at the limiting pole, after decreasing r_B once the theorem supplies a single analytic branch w_n

on every $U_\ell^\pm$, uniformly for all sufficiently large n, and these branches converge uniformly there.

Choose positive constants c_B and C_B, independent of n, with the following property. For each corresponding finite pole, set

$$\nu_{\ell,n}^\pm = np_{\ell,n}^\pm.$$

One may choose the local branch so that $w_n(\nu_{\ell,n}^\pm) = 2\pi\mathrm{i}\ell$ and

$$\left|w_n'(\nu_{\ell,n}^\pm)\right| \ge 2c_B, \qquad \sup_{|\nu-\nu_{\ell,n}^\pm|\le r_B} |w_n''(\nu)| \le C_B.$$

The first bound follows from the displayed derivative and the simple limiting pole. Cauchy's estimate on the doubled disks gives the second. Decrease r_B so that $C_B r_B/2 \le c_B$. Taylor's formula then gives

$$|w_n(\nu) - 2\pi\mathrm{i}\ell| \ge c_B|\nu - \nu_{\ell,n}^\pm| \qquad (|\nu - \nu_{\ell,n}^\pm| \le r_B).$$

Outside the deleted An^{-2} disk this is at least $c_B A/n$. On the complement of those fixed scaled neighborhoods inside the compact ν-set just obtained, the limiting branch has positive distance from the corresponding representative $2\pi\mathrm{i}\ell$. Uniform convergence and compactness therefore give a positive lower bound for the finite-n distance. The reciprocal contribution of all these finitely many indices is thus at most $C_B(1 + n/A)$.

Coalescing first pair. At $\alpha = q_\star$ and $\ell = 1$, write $\Delta = \lambda - \lambda_c$. The two first poles are $\lambda_c \pm r_n$, with $r_n = O(n^{-2})$. Outside the grouped disk,

$$|D_1(\lambda)| = |\Delta^2 - r_n^2| \ge cA^2 n^{-3}.$$

Since $|\sin(q_\star/n)| \asymp n^{-1}$, (7.16) implies $|\theta| \ge cA^2/n \ge cA/n$. This supplies the same reciprocal bound as a simple pole without separating the coalescing pair.

Remaining wave numbers. It remains to treat $\ell \ge L+1$. Choose $q_0 \in (0, \pi/2)$. When $q \le q_0$, the two poles are conjugate for large n; write them as $h_{\ell,n} \pm \mathrm{i}t_{\ell,n}$. Every point of $\Omega_{B,n}$ has horizontal separation at least

$$u_{\ell,n} = 1 - \cos(\ell q_\star/n) - B/n^2$$

from their common real part. The midpoint definition of B and $1 - \cos q = q^2/2 + O(q^4)$ give

$$u_{\ell,n} \ge u_{L+1,n} \ge c\sqrt{B}\, n^{-2}, \qquad t_{\ell,n} \ge cq.$$

For $u \ge 0$ and real y, t,

$$|(u + \mathrm{i}(y-t))(u + \mathrm{i}(y+t))| \ge 2u|t|, \tag{7.17}$$

because the difference of the squares of the two sides is $(u^2 + y^2 - t^2)^2$. Hence $|D_\ell(\lambda)| \ge cu_{\ell,n}q$. Since $q \ge 2q_\star/n$ and $|\theta| < 1$, the right side of (7.16) is at most $Cq|\theta|/n$. Therefore

$$|\theta| \ge cnu_{\ell,n} \ge c\sqrt{B}/n.$$

When $q \in [q_0, \pi]$, both poles have real part at most $-c(q_0) < 0$, while the boundary of $\Omega_{B,n}$ tends to zero. Thus $|D_\ell(\lambda)| \geq c(q_0)^2$, which is incompatible with (7.16) and $|\theta| < 1$ for large n. This proves (7.13).

Supercritical window. Now suppose $\alpha > q_\star$. The choice $d_1 < d_{\rm cut} < d_2$ places only the slow first pole in $\Omega_{{\rm cut},n}$. Put

$$d_{\rm f} = \alpha + \sqrt{\alpha^2 - q_\star^2};$$

the scaled slow and fast first poles converge to $-d_1$ and $-d_{\rm f}$, respectively, and $d_{\rm f} > \alpha > d_{\rm cut}$ because $d_2 \leq \alpha$. The simple-pole inverse-function argument used above gives $|\theta| \geq cA/n$ in a fixed scaled neighborhood of the slow pole after its disk is deleted.

It remains to justify a uniform bound on the complement of that neighborhood. If $|\theta| < 1$, then (7.16) and the factorization (7.15) give

$$\min_{\sigma\in\{+,-\}} |\lambda - p_{1,n}^\sigma|^2 \leq |D_1(\lambda)| \leq Cn^{-2}.$$

Since $p_{1,n}^\pm = O(n^{-1})$, the scaled variable $\nu = n\lambda$ therefore lies in a fixed disk $|\nu| \leq M$. Fix a sufficiently small $r > 0$ and consider the compact set

$$K_r = \{\nu : \Re\nu \geq -d_{\rm cut},\ |\nu| \leq M,\ |\nu + d_1| \geq r\}.$$

There is a positive lower bound for $|\theta|$ over the corresponding finite-n points. Otherwise, along a subsequence one would have $\theta \to 0$ and $\nu \to \nu_\infty \in K_r$. Multiplying (7.16) by n^2 gives $n^2 D_1(\nu/n) \to 0$, whereas the scaled factorization gives, uniformly on this compact set,

$$n^2 D_1(\nu/n) \longrightarrow (\nu_\infty + d_1)(\nu_\infty + d_{\rm f}).$$

Hence $\nu_\infty = -d_1$ or $-d_{\rm f}$. The first alternative is excluded by the definition of K_r, and the second by $d_{\rm f} > d_{\rm cut}$. Compactness therefore yields $|\theta| \geq c$ on this complement, uniformly for all sufficiently large n.

Choose L_0 with $L_0 q_\star > 2\alpha$. For each fixed $2 \leq \ell \leq L_0$, both pole factors have horizontal distance at least c/n from $\Omega_{{\rm cut},n}$; hence $|D_\ell(\lambda)| \geq cn^{-2}$. Equation (7.16) gives $|D_\ell(\lambda)| \leq C|\theta|n^{-2}$ when $|\theta| < 1$, so $|\theta| \geq c$. This also covers a coalescence $\alpha = \ell q_\star$.

For $\ell > L_0$ and $q \leq q_0$, the poles are conjugate, $t_{\ell,n} \geq cq$, and their horizontal separation from the window is at least

$$u_{\ell,n} = \frac{\alpha - d_{\rm cut}}{n} + 1 - \cos q.$$

Equations (7.17) and (7.16) give

$$|\theta| \geq cnu_{\ell,n} \geq c(\alpha - d_{\rm cut}).$$

The fixed separation for $q \in [q_0, \pi]$ again rules out $|\theta| < 1$. Combining the slow-pole estimate with these fixed lower bounds proves (7.14). ∎

Lemma 7.5 *(Critical off-pole Green estimate)***.** Fix $\alpha > 0$ and $R > 0$, put $\omega = \alpha/n$ and $q_\star = 2\pi$, and restrict λ to $|\lambda| \leq R$. The following estimates hold.

1. Suppose $\alpha \le q_\star$. For an integer $L \ge 1$, set

$$B = \frac{q_\star^2}{4}\{L^2 + (L+1)^2\}, \qquad \Omega_{B,n} = \{\Re\lambda \ge -\alpha/n - B/n^2\}.$$

   Delete the disks of radius An^{-2} about all simple homogeneous poles in $\Omega_{B,n}$. If $\alpha = q_\star$, replace the two disks at the first poles by the disk of radius $An^{-3/2}$ about their midpoint. There are constants $C = C(\alpha, R)$ and $C_B = C_B(\alpha, B, R)$, independent of A and n, such that, for every fixed $A \ge 1$, the remaining set satisfies the following bound whenever $n \ge N(\alpha, B, R, A)$:

$$|G_n(\lambda)| \le C_B\left(1 + \frac{n}{A}\right) + C\frac{n}{\sqrt{B}}. \tag{7.18}$$

2. Suppose $\alpha > q_\star$. Define

$$d_1 = \alpha - \sqrt{\alpha^2 - q_\star^2}, \qquad d_2 = \alpha - \Re\sqrt{\alpha^2 - 4q_\star^2},$$

   choose $d_{\rm cut} \in (d_1, d_2)$, and put $\Omega_{{\rm cut},n} = \{\Re\lambda \ge -d_{\rm cut}/n\}$. After deleting the disk of radius An^{-2} about the slow first pole, one has

$$|G_n(\lambda)| \le C_{\rm cut}\left(1 + \frac{n}{A}\right) \tag{7.19}$$

   throughout $\Omega_{{\rm cut},n} \cap \{|\lambda| \le R\}$. Here $C_{\rm cut} = C_{\rm cut}(\alpha, d_{\rm cut}, R)$ is independent of A and n, and, for every fixed $A \ge 1$, the estimate holds for $n \ge N(\alpha, d_{\rm cut}, R, A)$.

Proof. Write $w = n \operatorname{arccosh} x$ and $\Theta = (\lambda + 1 - x)/s$. Besides (5.15), we use the exact identity

$$\Theta^2 = \frac{\lambda(\lambda+2)}{(\lambda+2\omega)(\lambda+2+2\omega)}. \tag{7.20}$$

It follows by squaring the second identity in (6.14) and using

$$x + 1 = \frac{(\lambda+2)(\lambda+2+2\omega)}{2a}.$$

A branch change replaces w by $\pm w + 2\pi i n m$. Under the sign change both $s = \sinh(w/n)$ and $\coth(w/2)$ change sign, while the additive term is a period. Thus $\Theta\coth(w/2)$ is unchanged, as are the distances in Lemma 7.4.

The right-edge windows stay a fixed multiple of n^{-1} to the right of the tumble pole -2ω and a fixed distance to the right of $-2-2\omega$. Consequently $|a|$ is bounded below and (7.20) gives $|\Theta| \le C$ there. Fix a small $\eta > 0$. If $d_{0,n}(w) \le \eta$, a branch change puts $|w| \le \eta$. Taking the quotient of the two identities in (6.14) where it is defined, extending through removable points by analytic continuation, and using $(\cosh t - 1)/\sinh t = \tanh(t/2)$ gives the exact factorization

$$\Theta\coth(w/2) = \frac{\lambda+2}{\lambda+2\omega}\tanh\!\left(\frac{w}{2n}\right)\coth(w/2). \tag{7.21}$$

In the subcritical window,

$$\Re(\lambda + 2\omega) \ge \frac{\alpha}{n} - \frac{B}{n^2} \ge \frac{\alpha}{2n}$$

for all sufficiently large n. In the supercritical window,

$$\Re(\lambda + 2\omega) \;\geq\; \frac{2\alpha - d_{\mathrm{cut}}}{n},$$

where $2\alpha - d_{\mathrm{cut}} > 0$ because $d_{\mathrm{cut}} < d_2 \leq \alpha$. Since also $|\lambda + 2\omega| \geq |\Im\lambda|$, the two cases give

$$|\lambda + 2\omega| \;\geq\; c(n^{-1} + |\Im\lambda|). \tag{7.22}$$

Since $|w| \leq \eta$, the first identity in (6.14) and the upper bound on $|a|$ imply $|\lambda(\lambda + 2\omega)| \leq Cn^{-2}$, and hence $|\lambda| \leq Cn^{-1}$. Indeed, if $|\lambda| \leq 4\omega = 4\alpha/n$, this is immediate; otherwise $|\lambda + 2\omega| \geq |\lambda| - 2\omega \geq |\lambda|/2$, and therefore $|\lambda|^2 \leq 2Cn^{-2}$. The lower bound on $|a|$ and the same identity then give

$$|w|^2 \;\leq\; Cn^2|\lambda|\,|\lambda + 2\omega| \;\leq\; Cn|\lambda + 2\omega|.$$

Finally, Taylor's theorem gives, uniformly for $n \geq 1$ and $|w| \leq \eta$,

$$n \tanh\!\Big(\frac{w}{2n}\Big) \coth(w/2) \;=\; 1 + \mathcal{E}_n(w), \qquad |\mathcal{E}_n(w)| \;\leq\; C|w|^2.$$

Moreover, the preceding bound on $|w|^2$ gives

$$\left|\frac{\lambda + 2}{n(\lambda + 2\omega)}\mathcal{E}_n(w)\right| \;\leq\; \frac{C_R|w|^2}{n|\lambda + 2\omega|} \;\leq\; C_R.$$

Substitution in (7.21) therefore gives

$$\Theta \coth(w/2) \;=\; \frac{\lambda + 2}{n(\lambda + 2\omega)} + O(1), \tag{7.23}$$

uniformly in either window. Equation (7.22) shows that the leading term, and hence the entire expression, is bounded.

If $d_{0,n}(w) > \eta$, then

$$\mathrm{dist}(w, 2\pi \mathrm{i}\mathbb{Z}) \;=\; \min\{d_{0,n}(w), d_{*,n}(w)\}.$$

The finite Green formula and (6.18) therefore give, throughout this complementary region,

$$|G_n(\lambda)| \;\leq\; C\left\{1 + d_{*,n}(w)^{-1}\right\}. \tag{7.24}$$

Equation (7.23) bounds the first region. On the second region, combine (7.24) with item 1 or item 2 of Lemma 7.4. This gives respectively (7.18) and (7.19), with exactly the stated dependence of the constants. ■

7.4. Root counts and right-edge exhaustion

Lemma 7.6 *(Critical pole-neighborhood root counts)***.** Set $\omega = \alpha/n$ and $\omega_0 = \beta/n$, where $\alpha > 0$ and $\beta \geq 0$ are fixed, put $q_\star = 2\pi$, and write $b = \beta - \alpha$.

1. Fix $\ell \geq 1$ with $\alpha \neq \ell q_\star$. There is $A_\ell > 0$ such that, for each sign and every fixed $A \geq A_\ell$, all sufficiently large n have the property that the disk $|\lambda - p^\pm_{\ell,n}| < An^{-2}$ contains exactly one zero $\widehat{p}^\pm_{\ell,n}$ of $Q_{n,-}$, counted with algebraic multiplicity, and

$$\widehat{p}^\pm_{\ell,n} = p^\pm_{\ell,n} - 4bR^\pm_{\ell,n} n^{-2} + O(n^{-3}). \tag{7.25}$$

   The threshold in n may depend on A.

2. If $\alpha = q_\star$, put

$$\lambda_c = \cos(q_\star/n) - 1 - q_\star/n, \qquad r_n = \sqrt{(q_\star/n)^2 - \sin^2(q_\star/n)}.$$

   There is $A_* > 0$ such that, for every fixed $A \geq A_*$ and all sufficiently large n, the disk $|\lambda - \lambda_c| < An^{-3/2}$ contains exactly two zeros of $Q_{n,-}$, counted with algebraic multiplicity. The same two zeros occur for every such fixed enlargement A.

Proof. If $b = 0$, then $Q_{n,-} = H^{(0)}_{n,-}$. A simple-pole disk contains exactly its homogeneous root, so item 1 holds with zero shift. At $\alpha = q_\star$, the two first homogeneous roots are $\lambda_c \pm r_n$, where $r_n = O(n^{-2})$; both lie in every fixed $An^{-3/2}$ disk for all sufficiently large n, while all other homogeneous roots are separated on the n^{-1} scale. Thus item 2 also holds. Assume below that $b \neq 0$, so $\delta = b/n$.

At a simple pole, Lemma 7.3 gives

$$G_n(\lambda) = \frac{2R^\pm_{\ell,n}}{n(\lambda - p^\pm_{\ell,n})} + G^{\mathrm{reg}}_{\ell,n}(\lambda), \qquad G^{\mathrm{reg}}_{\ell,n} = O(1)$$

on a fixed scaled neighborhood. Multiplication of the secular equation by $\lambda - p^\pm_{\ell,n}$ gives

$$(\lambda - p^\pm_{\ell,n})(1 + 2\delta G_n) = (\lambda - p^\pm_{\ell,n})\left(1 + \frac{2b}{n} G^{\mathrm{reg}}_{\ell,n}\right) + \frac{4bR^\pm_{\ell,n}}{n^2}. \tag{7.26}$$

Compare this expression with

$$h_0(\lambda) = \lambda - p^\pm_{\ell,n} + 4bR^\pm_{\ell,n} n^{-2}.$$

If $\sigma_\ell = \sqrt{\alpha^2 - \ell^2 q_\star^2}$ with the branch inherited from Definition 7.2, then

$$R^\pm_{\ell,n} \longrightarrow \frac{1}{2} \mp \frac{\alpha}{2\sigma_\ell} \neq 0;$$

the inequality follows from $\ell \geq 1$. Thus the residues remain bounded and bounded away from zero. Choose A_ℓ so that the zero of h_0 lies a fixed relative distance inside the disk. On its boundary, the difference from h_0 is $O(n^{-3})$, whereas $|h_0| \geq c_A n^{-2}$. Rouché's theorem gives exactly one pole-cleared zero. At that zero, (7.26) first gives $\lambda - p^\pm_{\ell,n} = O(n^{-2})$ and then, after substitution in its regular-part term, gives the expansion (7.25).

On this disk write $H^{(0)}_{n,-} = (\lambda - p^\pm_{\ell,n})J_{\ell,n}$, where $J_{\ell,n}$ is nonvanishing. The residue $R^\pm_{\ell,n}$ is nonzero, and the right side of (7.26) is nonzero at the homogeneous pole. Thus that pole is canceled rather than retained, and the Rouché zero is exactly one zero of $Q_{n,-}$ with its algebraic multiplicity. This proves item 1.

Now let $\alpha = q_\star$ and put $\Delta = \lambda - \lambda_c$. By Lemma 7.3,

$$(\Delta^2 - r_n^2)(1 + 2\delta G_n) = (\Delta^2 - r_n^2)\left(1 + \frac{2b}{n}H_n\right) + \frac{4b}{n^2}(\Delta - q_\star/n), \tag{7.27}$$

where $H_n = O(1)$ throughout the grouped disk. Compare this with

$$P_{0,n}(\Delta) = \Delta^2 - r_n^2 + \frac{4b}{n^2}(\Delta - q_\star/n).$$

With $\Delta = n^{-3/2}Y$, one has locally uniformly in Y

$$n^3 P_{0,n}(n^{-3/2}Y) = Y^2 - 4bq_\star + O(n^{-1/2}).$$

Choose $A_* > 2\sqrt{|b|q_\star}$. Rouché's theorem applied to this scaled quadratic shows that both zeros of $P_{0,n}$ lie a fixed relative distance inside $|\Delta| < A_* n^{-3/2}$. On the boundary of every fixed enlargement, the difference in (7.27) is $O(A^2 n^{-4})$, whereas $|P_{0,n}| \geq c_A n^{-3}$. Rouché's theorem gives exactly two pole-cleared zeros. At the homogeneous poles $\Delta = \pm r_n$, the right side of (7.27) equals

$$\frac{4b}{n^2}(\pm r_n - q_\star/n),$$

which is nonzero for all sufficiently large n, because $r_n = q_\star^2/(\sqrt{3}\,n^2) + O(n^{-4})$. The factor $\Delta^2 - r_n^2$ is the complete homogeneous factor on the disk and the remaining factor of $H_{n,-}^{(0)}$ is nonvanishing. Hence the two Rouché zeros are exactly the two zeros of $Q_{n,-}$ there. Applying the count once at A_* and once at any fixed $A \geq A_*$ shows that the annular difference contains no zero, proving item 2. ∎

Lemma 7.7 *(Critical zero-wave affected root)***.** Set $\omega = \alpha/n$ and $\omega_0 = \beta/n$, where $\alpha > 0$ and $\beta \geq 0$ are fixed, and put $b = \beta - \alpha$. For all sufficiently large n, the affected polynomial $Q_{n,-}$ has exactly one zero in a fixed $O(n^{-3})$ neighborhood of $-2\alpha/n - 2b/n^2$. Counted with algebraic multiplicity, it is

$$\lambda_{0,-} = -\frac{2\alpha}{n} - \frac{2b}{n^2} + O(n^{-3}). \tag{7.28}$$

If $b \neq 0$, the homogeneous pole $-2\alpha/n$ is canceled; if $b = 0$, the displayed zero is that homogeneous endpoint root.

Proof. Write $\varepsilon = \lambda + 2\alpha/n$. Equation (7.10) turns the pole-cleared secular equation into

$$\varepsilon\left(1 + \frac{2b}{n}K_n(\lambda)\right) + \frac{2b}{n^2} = 0. \tag{7.29}$$

Let C_0 be the bound for K_n. On $|\varepsilon + 2b/n^2| = Rn^{-3}$, the difference between the left side and $\varepsilon + 2b/n^2$ is at most

$$\frac{2|b|C_0}{n}\left(\frac{2|b|}{n^2} + \frac{R}{n^3}\right) \leq C_1 n^{-3}$$

for $n \geq R$, where C_1 is independent of R. Choose $R > C_1$ and apply Rouché's theorem to obtain the unique zero and (7.28).

On the endpoint disk write $H^{(0)}_{n,-}(\lambda) = \varepsilon J_n(\lambda)$, with J_n holomorphic and nonvanishing. Since $G_n = 1/(n\varepsilon) + K_n$ and $\delta = b/n$,

$$Q_{n,-}(\lambda) = J_n(\lambda)\left[\varepsilon\left(1 + \frac{2b}{n}K_n(\lambda)\right) + \frac{2b}{n^2}\right].$$

For $b \neq 0$ the bracket is nonzero at $\varepsilon = 0$, so the homogeneous pole is canceled. For $b = 0$ the bracket is ε, so the endpoint root is retained. In either case the counted zero is the unique zero of $Q_{n,-}$ in the disk. ■

Definition 7.8 (First-pole candidate multiset)**.** Fix $\alpha > 0$ and $\beta \geq 0$, set $\omega = \alpha/n$, $\omega_0 = \beta/n$, and put $q_\star = 2\pi$. For a fixed $A > 0$, let $\mathcal{C}_{1,n}(A)$ contain, with algebraic multiplicity, the protected roots $p^{\pm}_{1,n}$ and, if $\alpha \neq q_\star$, the affected roots in the two disks $|\lambda - p^{\pm}_{1,n}| < An^{-2}$. If $\alpha = q_\star$, instead include all affected roots in the disk centered at $\lambda_c = \cos(q_\star/n) - 1 - q_\star/n$ given by

$$|\lambda - \lambda_c| < An^{-3/2}.$$

Lemma 7.9 *(Critical right-edge exhaustion)***.** Set $\omega = \alpha/n$ and $\omega_0 = \beta/n$, where $\alpha > 0$ and $\beta \geq 0$ are fixed, and put $q_\star = 2\pi$. There is $A_0 = A_0(\alpha,\beta) > 0$ such that, for every fixed $A \geq A_0$, $\mathcal{C}_{1,n}(A) = \mathcal{C}_{1,n}(A_0)$ for all sufficiently large n. Every nonzero eigenvalue with maximal real part belongs to $\mathcal{C}_{1,n}(A_0)$. More precisely, if $\Lambda_{1,n}$ is the largest real part within that multiset, then every nonzero eigenvalue outside it satisfies

$$\Re\lambda \leq \Lambda_{1,n} - \begin{cases} cn^{-2}, & \alpha \leq q_\star, \\ cn^{-1}, & \alpha > q_\star, \end{cases} \tag{7.30}$$

for a constant $c = c(\alpha,\beta) > 0$.

Proof. Right-edge secular exclusion. The protected roots are the homogeneous roots from Definition 7.2. Put $b = \beta - \alpha$. The argument below includes $b = 0$; in that case the affected roots coincide with their homogeneous odd-sector counterparts, and Lemmas 7.6 and 7.7 give the same local counts with zero shift.

Gershgorin's theorem confines all eigenvalues to a fixed disk; choose R so that this disk lies in $|\lambda| \leq R$ and use Lemma 7.5 with that R. The constants are chosen in the following order.

1. If $\alpha \leq q_\star$, choose the integer L in Lemma 7.5 sufficiently large and set $B = q_\star^2\{L^2 + (L+1)^2\}/4$. Increase L, if necessary, until the protected first root lies inside $\Omega_{B,n}$ with an n^{-2}-scale margin and $2|b|C/\sqrt{B} < 1/8$. If $\alpha > q_\star$, fix once and for all $d_{\mathrm{cut}} \in (d_1, d_2)$ and use $\Omega_{\mathrm{cut},n}$.
2. Choose A_0 so that $2|b|C_B/A_0 < 1/8$ in the first case or $2|b|C_{\mathrm{cut}}/A_0 < 1/4$ in the second. Increase A_0, if necessary, so that it exceeds the finitely many simple-pole thresholds and, when $\alpha = q_\star$, the grouped threshold in Lemma 7.6. When $\alpha > q_\star$, include the threshold for the fast first pole as well.

3. Fix any $A \geq A_0$. Only now choose n large enough for the estimates with this A and, additionally, for $2|b|C_B/n < 1/4$ or $2|b|C_{\mathrm{cut}}/n < 1/4$, respectively.

Since $2\delta = 2b/n$, item 1 of the off-pole estimate gives explicitly

$$|2\delta G_n| \leq \frac{2|b|C_B}{n} + \frac{2|b|C_B}{A} + \frac{2|b|C}{\sqrt{B}} < \frac{1}{2}$$

when $\alpha \leq q_\star$. Item 2 gives

$$|2\delta G_n| \leq \frac{2|b|C_{\mathrm{cut}}}{n} + \frac{2|b|C_{\mathrm{cut}}}{A} < \frac{1}{2}$$

when $\alpha > q_\star$. Thus in either case

$$|2\delta G_n(\lambda)| < \frac{1}{2}. \tag{7.31}$$

Thus every affected zero in the chosen right-edge window lies in one of the pole neighborhoods specified in that lemma.

Local counts and candidate stabilization. Apply Lemma 7.6 to the finitely many simple pole neighborhoods in the chosen window. Each contains exactly one affected root, with expansion (7.25). When $\alpha = q_\star$, apply its grouped assertion to the first neighborhood; together with (7.31), these counts account for every affected root in the window. When $\alpha > q_\star$, the window contains only the slow first pole, whereas Definition 7.8 also contains the fast first disk. The simple-pole assertion of the same lemma supplies its unique affected root.

For $\alpha \neq q_\star$, applying the simple-pole count once with A_0 and once with A shows that the annular difference between the two first disks contains no affected root. For $\alpha = q_\star$, the corresponding statement is part of the grouped count. The protected roots do not depend on A, and hence $\mathcal{C}_{1,n}(A) = \mathcal{C}_{1,n}(A_0)$ with algebraic multiplicity. This proves the asserted stabilization.

Subcritical case. Whenever $\ell \geq 1$ is fixed and $\alpha < \ell q_\star$, the two poles are nonreal conjugates and

$$\Re R^{\pm}_{\ell,n} = \frac{1}{2}, \qquad \Re p^{\pm}_{\ell,n} = -\frac{\alpha}{n} - \frac{\ell^2 q_\star^2}{2n^2} + O(n^{-4}). \tag{7.32}$$

Suppose first that $\alpha < q_\star$. Then equations (7.25)– (7.32) show that the protected-versus-affected real-part comparison is the same for every fixed ℓ, and that the $\ell = 2$ cluster is to the left of the first by $3q_\star^2/(2n^2) + O(n^{-3})$. We now make the global partition explicit. The midpoint definition of B in Lemma 7.4 puts precisely the fixed-index nonzero-wave poles with $1 \leq \ell \leq L$ in $\Omega_{B,n}$ for all large n. Lemma 7.6 gives one affected root at each of their simple poles, while (7.31) leaves no affected root on the complement of those disks. Among the finitely many clusters with $2 \leq \ell \leq L$, the preceding comparison puts every selected root at least $c_1 n^{-2}$ to the left of the selected $\ell = 1$ cluster. Every remaining nonzero-wave root is outside $\Omega_{B,n}$; because the protected first root lies inside that window with the margin fixed in item 1 of the constant choice, such a root is at least $c_2 n^{-2}$ to the left of the largest real part in the $\ell = 1$ cluster. Thus the nonzero-wave roots outside the $\ell = 1$ cluster satisfy (7.30) with $c = \min(c_1, c_2)$.

For every fixed $\alpha > 0$, the zero-wave endpoint is the only unpaired affected pole on the $O(n^{-1})$ scale; the $q = \pi$ endpoint, when present, stays a fixed distance into the left half-plane. Lemma 7.7 gives its unique affected root and places its real part at $-2\alpha/n + O(n^{-2})$. For $\alpha < q_\star$ this is a distance of order n^{-1} to the left of the $\ell = 1$ cluster. Combined with the nonzero-wave partition above, this proves (7.30) for $\alpha < q_\star$.

Critical case. At $\alpha = q_\star$, the first protected cluster contains

$$-\frac{q_\star}{n} + \left(-\frac{q_\star^2}{2} + \frac{q_\star^2}{\sqrt{3}}\right) n^{-2} + O(n^{-3}). \tag{7.33}$$

For every fixed $\ell \geq 2$, equations (7.25)– (7.32) apply with $\alpha = q_\star$. If $\beta \leq q_\star$, its largest possible real part is below (7.33) by

$$\left[\frac{(\ell^2-1)q_\star^2}{2} + \frac{q_\star^2}{\sqrt{3}} - 2(q_\star - \beta)\right] n^{-2} + O(n^{-3}),$$

which is positive because $q_\star > 2\sqrt{3}$. If $\beta > q_\star$, the affected higher modes move to the left and the protected comparison is even stronger. Estimate (7.31) accounts for all remaining affected roots. The endpoint formula (7.28) from Lemma 7.7 puts the zero-wave affected root at $-2q_\star/n + O(n^{-2})$ and hence a distance of order n^{-1} to the left of the $\ell = 1$ cluster. This proves the assertion at $\alpha = q_\star$.

Supercritical case. Finally, if $\alpha > q_\star$, the first slow pole has decay coefficient $d_1 = \alpha - \sqrt{\alpha^2 - q_\star^2}$. By the choice of d_{cut}, the only protected pole in $\Omega_{\mathrm{cut},n}$ is the first slow pole, and (7.31)– (7.25) show that its affected partner is the only affected root there. Both have real part $-d_1/n + O(n^{-2})$, whereas every nonzero root outside the window has real part at most $-d_{\mathrm{cut}}/n$. In particular, the endpoint root (7.28) is at $-2\alpha/n + O(n^{-2})$ and lies outside the window because $d_{\mathrm{cut}} < 2\alpha$. This gives a separation of order n^{-1} and completes the proof. ∎

Definition 7.10 (Critical first cluster)**.** Under the hypotheses of Lemma 7.9, choose $A_0 = A_0(\alpha, \beta)$ as in that lemma. The stabilized multiset $\mathcal{C}_{1,n}(A_0)$, independent of every fixed enlargement $A \geq A_0$ for all sufficiently large n, is the critical first cluster.

7.5. Proof of the bulk-critical theorem

Proof of Theorem 7.1. For $\alpha < q_\star$, the protected first pair has

$$\Re\lambda_1^+ = -\frac{\alpha}{n} - \frac{q_\star^2}{2n^2} + O(n^{-3}).$$

The pole residue tends to $R_1 = \frac{1}{2} + \mathrm{i}\alpha/(2\sqrt{q_\star^2 - \alpha^2})$, and $\delta = (\beta - \alpha)/n$. Hence the affected real part shifts by $-2(\beta - \alpha)n^{-2} + O(n^{-3})$ by (7.25). Taking the larger real part gives (7.2).

For $\alpha > q_\star$, put $\sigma_\star = \sqrt{\alpha^2 - q_\star^2}$. The protected rate is $(\alpha - \sigma_\star)n^{-1} + q_\star^2(2n^2)^{-1} + O(n^{-3})$, while $R_1 \to -(\alpha - \sigma_\star)/(2\sigma_\star)$. The same pole equation shifts the affected eigenvalue by $2(\beta - \alpha)(\alpha - \sigma_\star)/(\sigma_\star n^2) + O(n^{-3})$. Selection of the larger real part gives (7.3). Lemma 7.9 shows in both cases that no root outside the critical first cluster of Definition 7.10 can compete with the selected root.

At $\alpha = q_\star$, the exact lattice splitting of the homogeneous first block is

$$\sqrt{\omega^2 - \sin^2(q_\star/n)} \;=\; \frac{q_\star^2}{\sqrt{3}\,n^2} + O(n^{-4}),$$

which proves (7.4) when the protected root has maximal real part. Near the center of this block, put

$$\lambda_c \;=\; \cos(q_\star/n) - 1 - q_\star/n, \qquad r_n \;=\; \sqrt{(q_\star/n)^2 - \sin^2(q_\star/n)}, \qquad \Delta \;=\; \lambda - \lambda_c.$$

Lemma 7.3 gives the uniform identity

$$G_n(\lambda) \;=\; \frac{2(\Delta - q_\star/n)}{n(\Delta^2 - r_n^2)} + H_n(\lambda), \qquad |H_n(\lambda)| \;\le\; C, \tag{7.34}$$

throughout a fixed $n\Delta$-neighborhood of zero. In particular this bound is uniform on every contour used below.

Put $\tau = n^{-1/2}$ and scale $\Delta = n^{-3/2}Y$. For $\beta > q_\star$, put $b = \beta - q_\star$. Since

$$n^4 r_n^2 \;=\; \frac{q_\star^4}{3} + O(n^{-2}),$$

the secular equation, uniformly for Y in a compact set bounded away from zero, is

$$F_n(Y) \;=\; 1 - \frac{4q_\star b}{Y^2} + \frac{4b\tau}{Y} + O(\tau^2) \;=\; 0. \tag{7.35}$$

The limiting function has simple zeros $Y_\pm = \pm 2\sqrt{q_\star b}$. On $|Y - Y_\pm| = C\tau$, its modulus is bounded below by $cC\tau$, while the difference in (7.35) is at most $C_1\tau$. Choosing $C > C_1/c$, Rouché's theorem gives one zero in each disk and hence two affected roots

$$\Delta \;=\; \pm 2\sqrt{q_\star b}\, n^{-3/2} + O(n^{-2}).$$

The positive displacement gives (7.5).

For $\beta < q_\star$, put $h = q_\star - \beta > 0$. With the same scaling, the secular equation is

$$\widetilde{F}_n(Y) \;=\; 1 + \frac{4q_\star h}{Y^2} - \frac{4h\tau}{Y} + O(\tau^2) \;=\; 0. \tag{7.36}$$

Its two leading zeros are $Y_\pm = \pm 2\mathrm{i}\sqrt{q_\star h}$. Substitution shows that $Y_\pm + 2h\tau$ has residual $O(\tau^2)$. On the circles $|Y - (Y_\pm + 2h\tau)| = C\tau^2$, the affine part has modulus at least $cC\tau^2$ and the remaining error is at most $C_1\tau^2$. A second application of Rouché's theorem therefore gives the affected pair

$$\Delta \;=\; \pm 2\mathrm{i}\sqrt{q_\star h}\, n^{-3/2} + 2hn^{-2} + O(n^{-5/2}). \tag{7.37}$$

Its decay rate exceeds the protected decay rate by

$$\left(\frac{q_\star^2}{\sqrt{3}} - 2h\right) n^{-2} + O(n^{-5/2}).$$

This coefficient is positive for $0 < h \leq q_\star$ because $q_\star = 2\pi > 2\sqrt{3}$. The protected root therefore has maximal real part for every $\beta < q_\star$; at $\beta = q_\star$ the two parity roots coincide. This proves (7.4); the case $\beta > q_\star$ was obtained from the positive displacement above. Lemma 7.9 supplies the required exclusion of every higher, endpoint, and off-pole affected root in both cases. ∎

8. Telegraph comparison

8.1. Homogeneous continuum approximations

The standard one-dimensional telegraph system follows from the right- and left-moving continuum equations in [2, Sec. II, Eqs. (2.2)–(2.4)]. In that convention each collision retains or reverses direction with equal probability, so a collision rate 2ω produces an actual reversal rate ω. The direct run-and-tumble derivation of the telegrapher equation is also given in [23, Sec. 2, Eqs. (1)–(7)]. Setting the speed to one gives the normalization below. We compare it with the next lattice correction before introducing an independently defined continuum point interaction at the same unit site spacing. No convergence of lattice operators under a vanishing-spacing limit is claimed.

Definition 8.1 (Two continuum approximations)**.** Let ρ denote total density and m polarization. Both fields are n-periodic in x. For a homogeneous ring of circumference n, the standard telegraph system is

$$\partial_t \rho = -\partial_x m, \qquad \partial_t m = -\partial_x \rho - 2\omega m. \tag{8.1}$$

Its slow Fourier eigenvalue at wave number q is

$$\lambda^+_{\mathrm{tel}}(q) = -\omega + \sqrt{\omega^2 - q^2}. \tag{8.2}$$

Here and below the square root has nonnegative real part and, when it is purely imaginary, nonnegative imaginary part. The lattice-diffusion-corrected system is

$$\partial_t \rho = -\partial_x m + \tfrac{1}{2}\partial_x^2 \rho, \qquad \partial_t m = -\partial_x \rho + \tfrac{1}{2}\partial_x^2 m - 2\omega m. \tag{8.3}$$

Its slow eigenvalue is

$$\lambda^+_{\mathrm{td}}(q) = -\frac{q^2}{2} - \omega + \sqrt{\omega^2 - q^2}. \tag{8.4}$$

Lemma 8.2 *(Second-order lattice expansion)***.** Let $\rho, m \in C^4(\mathbb{R}/n\mathbb{Z})$ and sample $u_j^\pm = \{\rho(j) \pm m(j)\}/2$. The homogeneous lattice generator induces

$$\begin{aligned} \dot\rho_j &= -m'(j) + \tfrac{1}{2}\rho''(j) + r_{\rho,j}, \\ \dot m_j &= -\rho'(j) + \tfrac{1}{2}m''(j) - 2\omega m(j) + r_{m,j}, \end{aligned} \tag{8.5}$$

where

$$|r_{\rho,j}| \le \tfrac{1}{6}\|m'''\|_\infty + \tfrac{1}{24}\|\rho^{(4)}\|_\infty,$$
$$|r_{m,j}| \le \tfrac{1}{6}\|\rho'''\|_\infty + \tfrac{1}{24}\|m^{(4)}\|_\infty. \tag{8.6}$$

Thus (8.3) is the second-order long-wave truncation of the lattice equations, with the remainder controlled in the sampled ℓ^∞ norm by (8.6). For a Fourier mode of wave number q, the centered first-difference error is of order q^3 and the centered second-difference error is of order q^4, making the long-wave meaning explicit.

Proof. Adding and subtracting the two homogeneous equations (2.1)–(2.2) gives the exact sampled identities

$$\dot\rho_j = \frac{\rho(j-1)+\rho(j+1)}{2} - \rho(j) + \frac{m(j-1)-m(j+1)}{2},$$
$$\dot m_j = \frac{\rho(j-1)-\rho(j+1)}{2} + \frac{m(j-1)+m(j+1)}{2} - m(j) - 2\omega m(j).$$

Taylor's theorem on the unit intervals adjacent to j bounds the error in each centered first difference by one sixth of the supremum of the third derivative, and the error in each centered second difference by one twenty-fourth of the supremum of the fourth derivative. Substitution gives (8.5)–(8.6). ■

Theorem 8.3 *(Homogeneous lattice versus telegraph spectral decay)***.** Fix $\omega > 0$, let $q = 2\pi/n$, and define $\gamma_{\rm tel} = -\Re\lambda^+_{\rm tel}(q)$ and $\gamma_{\rm td} = -\Re\lambda^+_{\rm td}(q)$. Then, as $n \to \infty$,

$$\gamma_n^{\rm hom} - \gamma_{\rm tel} = \frac{q^2}{2} - \left(\frac{1}{24} + \frac{1}{6\omega}\right) q^4 + O(q^6), \tag{8.7}$$

$$\gamma_n^{\rm hom} - \gamma_{\rm td} = -\left(\frac{1}{24} + \frac{1}{6\omega}\right) q^4 + O(q^6). \tag{8.8}$$

Consequently

$$\frac{\gamma_{\rm tel}}{\gamma_n^{\rm hom}} \longrightarrow \frac{1}{1+\omega}, \qquad \frac{\gamma_{\rm td}}{\gamma_n^{\rm hom}} \longrightarrow 1. \tag{8.9}$$

Separately, let $\alpha > 0$ be fixed and set $\omega = \alpha/n$. The standard telegraph decay rate is exactly

$$\gamma_{\rm tel} = \frac{\alpha - \Re\sqrt{\alpha^2 - (2\pi)^2}}{n}, \tag{8.10}$$

which agrees with the coefficient of n^{-1} in (7.2)–(7.5). For fixed $\beta \ge 0$, the defect-dependent corrections under $\omega_0 = \beta/n$, of orders n^{-2} and $n^{-3/2}$ in those formulas, are beyond this homogeneous telegraph comparison.

Proof. The exact lattice dispersion replaces $-q^2/2$ by $\cos q - 1$ and q inside the square root by $\sin q$. Expanding $\cos q$, $\sin^2 q$, and the square root gives (8.7)–(8.8). Dividing by the leading exact rate $(1+\omega)q^2/(2\omega)$ gives (8.9). Substitution of $\omega = \alpha/n$ and $q = 2\pi/n$ into (8.2), followed by taking the negative real part, gives (8.10). ■

8.2. Point-defect telegraph operator

For the Schrödinger operator, a one-dimensional point interaction can be encoded by continuity at the interaction point and a jump in the derivative; see [24, Part I, Ch. I.3, Eq. (3.1.9)]. After elimination of ρ from the eigenvalue equations below, the domain condition produces an analogous derivative jump for m. The operator here is nevertheless a first-order kinetic system, not a Schrödinger operator or a vanishing-spacing limit of the lattice generator.

Definition 8.4 (Point-defect telegraph operator)**.** Let $\eta \in \mathbb{R}$. For the kinetic defect in Definition 2.1, one has $\eta = \omega_0 - \omega \geq -\omega$. Here the continuum coordinate uses the same unit site spacing as the lattice, so the circle has circumference n. The Dirac coupling has coefficient 2η and enters the polarization equation as $-2\eta\,\delta_0(x)m$; no further lattice-spacing limit or rescaling of η is imposed. On the circle cut at zero, define the point-defect telegraph operator by

$$\mathcal{A}_\eta(\rho, m) \,=\, (-m', -\rho' - 2\omega m) \quad \text{on } (0, n),$$

with domain consisting of $m, \rho \in H^1(0, n)$ having traces that satisfy

$$m(0+) \,=\, m(n-), \qquad \rho(0+) - \rho(n-) \,=\, -2\eta m(0). \tag{8.11}$$

This is the distributional realization obtained by adding $-2\eta\,\delta_0(x)m$ to the second equation in (8.1).

Proposition 8.5 *(Off-spectrum point-defect telegraph secular equation)***.** For the operator in Definition 8.4, the domain is dense and the operator is closed. Define the homogeneous telegraph spectrum on this circle by

$$\Sigma_{\mathrm{tel}}(n, \omega) \,=\, \left\{-\omega \pm \sqrt{\omega^2 - (2\pi\ell/n)^2} \,:\, \ell \in \mathbb{Z}\right\}.$$

For $\lambda \notin \Sigma_{\mathrm{tel}}(n, \omega)$, put

$$\kappa \,=\, \sqrt{\lambda(\lambda + 2\omega)}$$

and interpret $\kappa^{-1}\coth(n\kappa/2)$ as the even meromorphic function of κ obtained by analytic continuation; no sign choice for the square root is required. The affected-parity eigenvalues outside $\Sigma_{\mathrm{tel}}(n, \omega)$ are exactly the zeros of

$$1 + \eta\frac{\lambda}{\kappa}\coth\left(\frac{n\kappa}{2}\right) \,=\, 0. \tag{8.12}$$

For the involution

$$\mathcal{R}(\rho, m)(x) \,=\, (\rho(n - x), -m(n - x)),$$

the reflection-even telegraph modes are unchanged. Formula (8.12) is the continuum analogue of (3.8) at unit site spacing; the exact lattice formula retains $1 - \cos q$ and $\sin q$, and (8.7)–(8.8) quantify the resulting spectral-decay error. At points of $\Sigma_{\mathrm{tel}}(n, \omega)$, this proposition identifies only the protected reflection-even family; it does not assert an affected-sector Jordan classification at homogeneous spectral points.

Proof. Domain and closedness. The jump in (8.11) cancels the delta distribution in $-\rho' - 2\eta\delta_0 m(0)$, while continuity of m prevents a delta distribution in $-m'$. Thus the stated domain gives the point interaction on $L^2((0,n);\mathbb{C}^2)$. It is densely defined because it contains $C_c^\infty((0,n);\mathbb{C}^2)$, and it is closed: the graph norm controls $\|m'\|_2$ directly and controls $\|\rho'\|_2$ by $\|\rho'\|_2 \le \|(\mathcal{A}_\eta(\rho,m))_2\|_2 + 2\omega\|m\|_2$; the trace theorem then preserves (8.11) under graph-norm limits.

Homogeneous spectrum. For $\eta = 0$, substitution of the periodic Fourier mode $e^{2\pi i\ell x/n}$ gives $\lambda(\lambda+2\omega) + (2\pi\ell/n)^2 = 0$. Hence its two roots, as ℓ ranges over $\mathbb{Z}$, are exactly $\Sigma_{\mathrm{tel}}(n,\omega)$. Indeed, the periodic Fourier transform identifies the homogeneous operator, with its periodic H^1 domain, with the orthogonal direct sum over $\ell \in \mathbb{Z}$ of the matrices

$$M_\ell := \begin{pmatrix} 0 & -iq_\ell \\ -iq_\ell & -2\omega \end{pmatrix}, \qquad q_\ell = \frac{2\pi\ell}{n}.$$

Their characteristic polynomials are $\lambda(\lambda+2\omega) + q_\ell^2$. To see that the direct sum adds no further spectrum, fix ζ outside this root set. Its inverse block is

$$(\zeta I - M_\ell)^{-1} = \frac{1}{\zeta(\zeta+2\omega)+q_\ell^2}\begin{pmatrix} \zeta+2\omega & -iq_\ell \\ -iq_\ell & \zeta \end{pmatrix}.$$

For all sufficiently large $|\ell|$, the denominator has modulus at least $q_\ell^2/2$, so $\|(\zeta I - M_\ell)^{-1}\| \le C_\zeta/(1+|q_\ell|)$; the finitely many remaining inverse blocks are bounded as well. Hence their direct sum maps L^2 boundedly into the weighted Fourier domain corresponding to periodic H^1, and is the resolvent of the homogeneous operator. Thus the displayed roots exhaust its spectrum.

Defect eigenfunctions. Let $\lambda \notin \Sigma_{\mathrm{tel}}(n,\omega)$. The eigenvalue equations away from zero are

$$-m' = \lambda\rho, \qquad -\rho' - 2\omega m = \lambda m.$$

Thus $\lambda \ne 0$ and elimination of ρ gives $m'' = \kappa^2 m$. Write

$$m(x) = A\cosh(\kappa(x-n/2)) + B\sinh(\kappa(x-n/2)).$$

The equality $m(0) = m(n)$ gives $B = 0$, since $\sinh(n\kappa/2) = 0$ would put λ in the homogeneous spectrum. Using $\rho = -m'/\lambda$, the second jump condition in (8.11) is equivalent to

$$m'(0) - m'(n) = 2\eta\lambda m(0). \tag{8.13}$$

Substitution of the even hyperbolic-cosine solution into (8.13) gives

$$-2\kappa\sinh(n\kappa/2) = 2\eta\lambda\cosh(n\kappa/2),$$

which is exactly (8.12). Conversely, if (8.12) holds outside the homogeneous spectrum, take $A = 1$, $B = 0$ in the displayed hyperbolic solution and put $\rho = -m'/\lambda$. Equation (8.13) and the continuity of m then give both conditions in (8.11), so this construction is a nonzero eigenfunction. Moreover $m(n-x) = m(x)$ and $\rho(n-x) = -\rho(x)$, hence $\mathcal{R}(\rho,m) = -(\rho,m)$ and the eigenfunction has affected parity.

Resolvent form and protected parity. Equivalently, the polarization entry of the homogeneous telegraph resolvent is

$$\frac{1}{n}\sum_{\ell\in\mathbb{Z}}\frac{\lambda}{\lambda(\lambda+2\omega)+(2\pi\ell/n)^2} = \frac{\lambda}{2\kappa}\coth\left(\frac{n\kappa}{2}\right).$$

The series converges normally away from the homogeneous spectrum and the last expression supplies its meromorphic continuation when κ is purely imaginary, in agreement with the direct jump calculation. Finally, a reflection-even vector satisfies $m(0+) = -m(n-)$; continuity of m therefore gives $m(0) = 0$. The jump vanishes, so these modes are protected exactly as in Lemma 4.2. ■

9. Conclusions

The rank-one polarization defect organizes the finite-ring problem into two complementary parts. The protected parity sector retains the homogeneous spectrum, while the affected sector is governed by one scalar Green function. This structure produces the derivative, Chebyshev, transfer, and parity forms of the characteristic polynomial and reduces the affected Jordan classification to finite polynomial root orders. The flat band at $\omega = 1$ is the singular exception, with its tuned J_3 block recorded separately.

On the infinite line, the same Green function distinguishes extended bulk spectrum from defect-localized modes. The infinite-support modes outside the bulk spectrum are described by the matching equations and their cubic elimination, and the singular transfer value supports a compact mode on three sites. The transfer alternatives exclude embedded ℓ^2 modes except at the singular flat band, whose eigenspace is given explicitly. Each simple infinite-support localized eigenvalue produces a finite-ring eigenvalue with exponentially small error and uniformly localized squared right-eigenvector amplitude.

The gap results separate two physical mechanisms. At fixed $\omega > 0$, reflection leaves one first-mode decay rate unchanged, so a slow site does not create a new scaling; a faster site changes only the n^{-3} correction. When both rates scale as n^{-1}, the gap itself is of order n^{-1}. The limiting first-mode exceptional scale $\alpha = 2\pi$ then reverses which side of the equal-rate line selects the affected parity mode and produces the $n^{-3/2}$ boundary correction for a faster defect.

The continuum comparison identifies why the leading telegraph truncation does not reproduce the fixed-rate lattice decay: unit-rate Poisson hopping contributes one half of the componentwise Laplacian, a second-order long-wave term absent from that truncation. Adding the term recovers the leading lattice rate, while the separately defined point-defect telegraph operator reproduces the parity protection and scalar secular structure at unit site spacing. Together, the finite, infinite, asymptotic, and continuum formulations expose the same protected/affected decomposition across every regime considered here.

Code availability

The manuscript source and the formula-audit code are available at https://github.com/skypher/kinetic-defect-ring; the repository history records the manuscript revision. From the repository root, the formula audit and the data generation for Figure 2 are run as

```
python3 -u scripts/verify_formulas.py,    python3 -u
                scripts/generate_figure_data.py.
```

The audit compares the characteristic, transfer, flat-band Jordan, localization, and representative gap formulas with direct finite-matrix or defining-equation calculations. Its scope and execution environment are documented in the repository; the computations are cross-checks rather than replacements for the arguments in the manuscript.

AI-use disclosure

OpenAI GPT-5.6 Sol (`gpt-5.6-sol`) assisted with developing and testing arguments, checking citations and calculations, writing verification code, and editing the exposition. The author directed the research, evaluated the final arguments, and assumes responsibility for the mathematical claims, references, computations, and text.

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